



DATA CENTER POWER PLANT CONTROL SYSTEM REQUIREMENTS

Document No.: ENG-STA-E-TBD

Rev: 2 – 26JAN26



REVISION HISTORY				
REVISION	DATE	AUTHOR	NATURE OF REVISION	APPROVAL
0	30SEP25	C. Gandy	Initial Release	L. Wise
1	01OCT25	C. Gandy	L. Wise Revisions	L. Wise
2	26JAN26	C. Gandy	Post CHIL Updates	

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1. EXECUTIVE SUMMARY

This requirements document defines a holistic Power Plant Control System for VoltaGrid’s typical Data Center Power Plant design. This document encompasses the full ecosystem of control: telemetric sensors, IEDs, electromechanical controls, manual backup control devices, Master Controllers, HMIs, Data processing servers/applications, operational networks, and integration strategies for AUX systems. This is in addition to a traditional PMS specification, that focuses only on master controllers and HMIs.

The intent is not for any single supplier to be responsible for the entire scope - instead this specification establishes the baseline requirements that should guide separate scopes of work across OEMs, EPCs, integrators, and internal VoltaGrid teams (including a PMS supplier) to ensure system components can integrate seamlessly into a unified, resilient plant control architecture.

Not all components of this specification will apply to every VoltaGrid Data Center Power Plant – each plant will include different makes, models, and quantities of generators and DERs as needed to satisfy the customer requirements. Additionally, the type and integration requirements for AUX systems, backup systems, customer control system interfaces, backup/utility power system connections, etc. may vary from plant to plant. The sections of this specification related to these systems are intended to communicate VoltaGrid’s high-level philosophical strategy for integrating these systems – not as prescriptive requirements for a single power plant.

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1.1. PHILOSOPHY

This specification focuses on three fundamental priorities:

Reliability and Resilience – The plant shall satisfy a minimum of 99.9% availability and shall be concurrently serviceable. In the event of a failure the plant shall remain operable, even if the control room or central automation layer is completely disabled. Individual gensets, DERs, breakers, and AUX systems shall be fully operable through their local IEDs and/or hardwired manual control devices. This ensures operators can still start, stop, and regulate equipment in the field even in a complete loss of remote control/automation.

No single point of failure - whether a master controller, HMI, server, generator, switch, conductor, etc. - should result in a plant-wide blackout. Furthermore, the plant should remain online and operable even in the event of a complete control room shut down (removing power from all redundant Master Controllers, HMIs, Data Aggregators, Network Switches, etc.). When controls/communications are reestablished, after an update, maintenance item, or failure, the control system shall reassume control without causing any equipment shutdowns, transient/step changes in any set points, or changes in operator-selected mode selections (Block, Auto, etc.).

Modular and Decoupled Design – Each subsystem (genset packages, electrical protection relays, gas distribution, SCR/urea systems, AUX MCCs, etc.) is required to operate independently, with open (non-proprietary, industry standard, and ethernet-based) communication protocols (Modbus, IEC 61850, IEEE C37.118, DNP3, MQTT).

This prevents vendor lock-in, simplifies upgrades, and enables straightforward integration of new assets such as BESS, new generator makes/models, or alternative DERs such as synchronous condensers.

Operator Accessibility – Operators shall always have multiple levels of control to ensure continuous plant operations:

1. Automatic control via Master Controllers
2. Remote control via HMIs, IEDs, and embedded subsystems
3. Manual Backup Control through switches, pushbuttons, and hand-operated valves

Master Controllers, HMIs, Data Aggregators, and Network architecture are required to be redundant to prevent loss of operator situational awareness.

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1.2. SCOPE

The document covers requirements across the **entire plant control system stack as shown in Figure 1:**

- **Plant Control System Objectives:** Performance, operability, functionality, capability, and reliability targets for the Plant Control System.
- **Telemetric Devices:** Sensors for pressure, temperature, vibration, flow, level, voltage, and current across gensets, distribution, gas, emissions, and AUX systems.
- **Electromechanical Control Devices:** Governors, exciters, breakers, contactors, valves, and drives, each with defined IED control interfaces.
- **Manual Backup Controls:** Local start/stop, selector switches, breaker operation switches, and emergency shutdown systems to guarantee control independent of automation layers.
- **IEDs and Controllers:** Genset controllers, engine controllers, electrical protection relays, AUX system controllers, and DER controllers - responsible for local machine-level automation while exposing all relevant data to the Plant Network and Cluster.
- **Network Infrastructure:** Redundant Ethernet-based communications backbone, supporting Modbus, IEC 61850, IEC C37.118, DNP, MQTT, etc.
- **Master Controllers:** Redundant supervisory controllers responsible for system-wide coordination (black start, bus transfer, load sharing, reserve margin management, emissions compliance). Failover logic ensures operations continue in droop mode if both controllers are lost.
- **HMIs and Data Aggregators:** Operator visibility into real-time status, alarms, trending data, and control functions. Fully redundant, with integration into cloud storage.
- **Gateway Servers, Historians, and Data Storage:** Long term data storage and view port population.
- **Test Reports:** RTDS Simulations, HIL FAT, and SAT

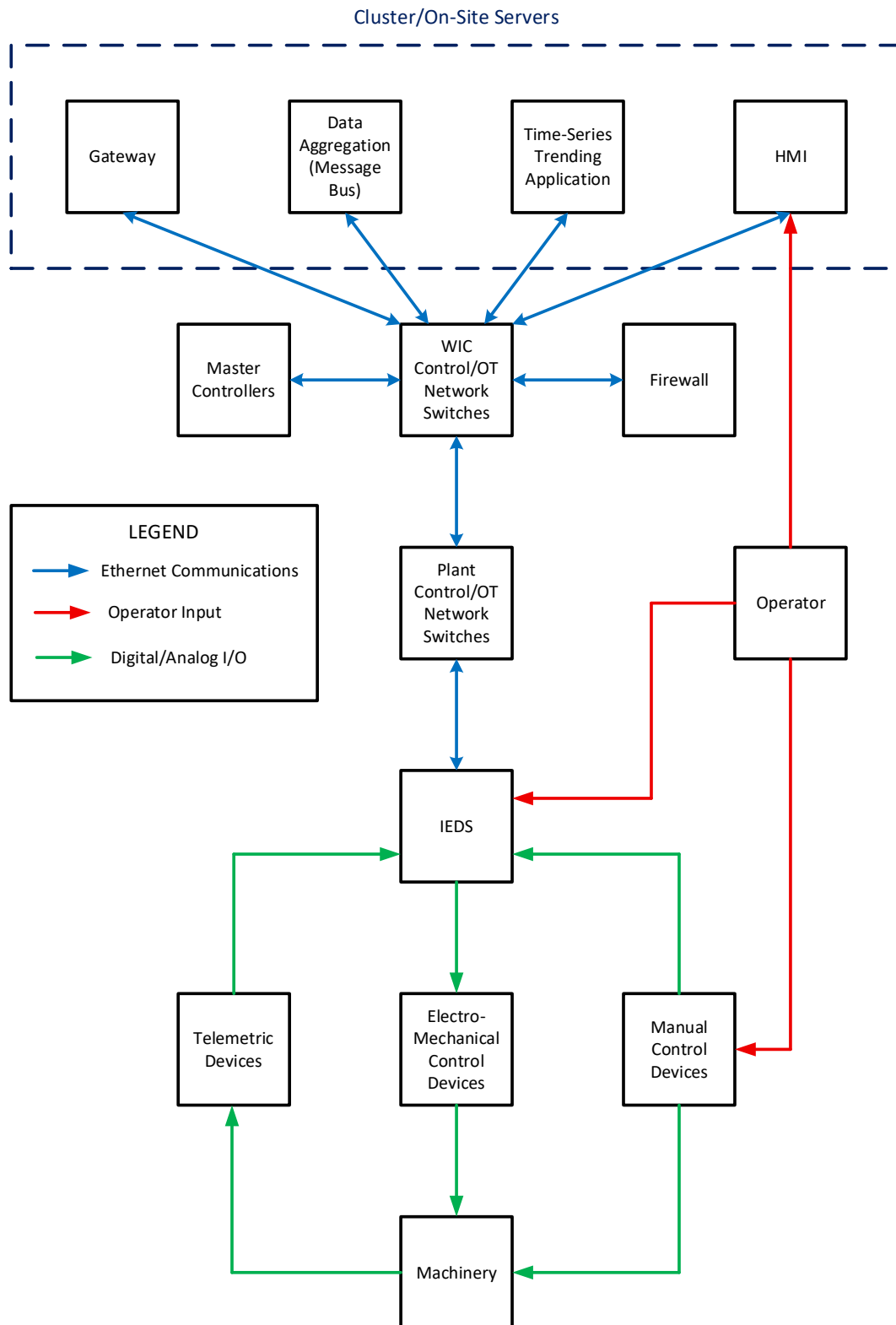


Figure 1: VoltaGrid Power Plant Control System Overview

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1.3. KEY FEATURES

- **Load Management:** Real and reactive power load sharing, reserve margin management, and minimum genset loading enforcement.
- **Failure Mode Resilience:** Loss of any single component (controller, HMI, network switch, aggregator) must not result in a system-wide blackout. Failures default to safe alarms, limp-mode, or limited unit shutdown rather than cascading outage.
- **Black Start & Dead Bus Arbitration:** Automated recovery sequences for plant restart following complete outage.
- **Fuel & Emissions Control:** Automated control of gas distribution and emissions treatment systems (SCR, bulk urea) independent of Master Controllers.
- **Testing & Documentation:** RTDS Simulation, hardware-in-the-loop testing, and SAT will validate all requirements. Each OEM/Supplier must provide cause-and-effect matrices, wiring diagrams, communication maps, and sequence of operation documentation to ensure consistency with the requirements defined here.

1.4. STRATEGIC VALUE

- **Resilient Operations:** Manual backup controls and robust controller redundancy ensures that the operator(s) can maintain power delivery in the field without reliance on the control room, aligning with VoltaGrid's ethos of toughness, practicality, and operational reliability.
- **Future-Proofed Architecture:** Modular, protocol-based design enables easy adoption of new technologies and DERs.
- **Integration Flexibility:** Requirements allow VoltaGrid to competitively source components without dependency on a single OEM, while ensuring compatibility.

1.5. CONCLUSION

This requirements specification defines a **resilient, flexible, and operator-friendly Plant Control System** tailored to VoltaGrid's mission-critical data center operations. By expanding beyond a narrow PMS approach, it ensures the system is robust against failures, accessible at multiple levels, and modular enough to integrate future technologies. For VoltaGrid and its customers, this means **assured uptime, simplified integration, and confidence that the plant remains under control even under worst-case conditions.**

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2. INTRODUCTION

To achieve safe, effective, efficient, reliable, and recordable operations; All modern power plants must include a robust and reliable Plant Control System. A modern Plant Control System typically consists of and/or relies on the following items:

- Telemetric devices including but not limited to:
 - Pressure sensors
 - Temperature Sensors
 - Status contacts
 - PTs
 - CTs
- Electromechanical control devices including but not limited to:
 - Governors (or pre-chamber gas valves)
 - Exciters
 - Trip Coils
 - Close Coils
 - Control Valves
 - Contactors
 - Visual indicators
- Manual control devices including but not limited to:
 - Buttons
 - Switches
 - Hand-operated valves
- IEDs including but not limited to:
 - Engine Controllers
 - Genset Controllers
 - DER Controllers for assets such as:
 - Synchronous Condensers
 - BESS
 - AUX system Controllers for systems including but not limited to:
 - SCRs
 - Bulk Urea Distribution System
 - Fuel distribution
 - Electrical Protection Relays

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- OT Network Including but not limited to:
 - Switches
 - Routers
 - Gateways
 - Firewalls
- Master Controller(s)
- Data Aggregator(s)
- HMI(s)
- Gateway Server(s)
- Historian(s)
- Time-Series Trending Application(s)

The purpose of this document is to define VoltaGrid’s requirements for the items listed above in Data Center Power Plant applications. The following sections provide the critical design criteria required for all stakeholders including system architects, Engineers, Operators, EPC’s, and vendors to develop control system components that meet VoltaGrid Data Center Power Plant Control System requirements.

It defines the scope, objectives, and key performance criteria for the system, ensuring alignment with business goals and technical requirements. This document serves as a foundation for subsequent design, development, and testing phases, fostering a unified approach to achieving operational excellence in data center power management.

3. OVERVIEW

A typical VoltaGrid Power Plant Control System Overview is shown below in **Figure 2**. In general, IEDs and/or the Master Controller actuate/modulate electromechanical control devices based on inputs/information from telemetric devices, manual control devices, and/or the operator (via the Control Room HMI). Meanwhile, the Data Aggregator collects and retransmits telemetric data and/or control inputs to the Control Room HMI, Time-Series Trending Application, and Gateway Server. The Control Room HMI provides the operator with limited control inputs as well as the data visualizations that are required for adequate situational awareness.

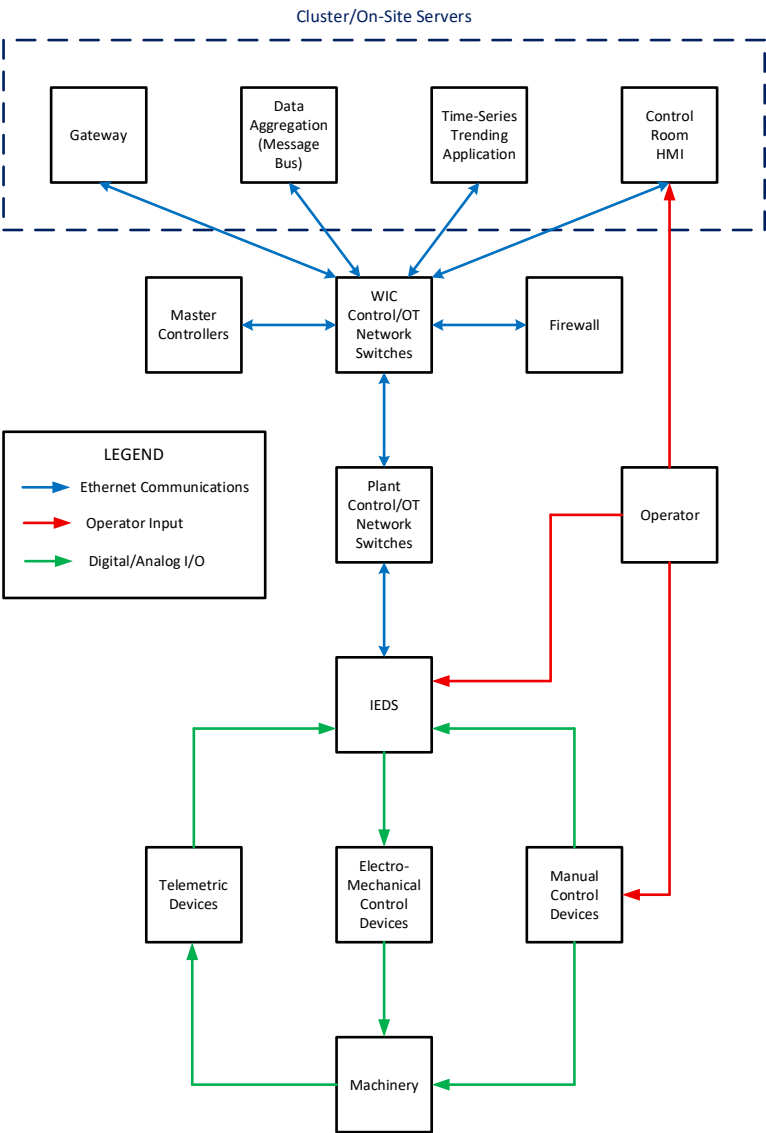


Figure 2: VoltaGrid Power Plant Control System Overview

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The “Machinery” that makes up a VoltaGrid Data Center Power Plant is divided into eight categories:

- MV Gensets
- DER Systems
- MV Electrical Distribution
- LV Electrical Distribution System
- Backup LV Gensets
- Fuel Gas Distribution System
- Bulk Urea Distribution System
- SCRs

Figure 3 shows a typical site layout for these assets. In this layout, piping from the Gas Yard and Bulk Urea Yard distributes fuel gas and Urea to the J620 (right) and J624 (left) gensets and their associated SCRs. This allows the engines to run in compliance with emissions regulations and produce mechanical power. The engines run at 1500 RPM and are coupled to their associated alternators through gear boxes allowing their 4-pole rotors to rotate at 1800RPM – producing 60Hz electric power at the stators.

Local genset excitation control systems maintain the stator voltage at a nominal 13.8kV phase-to-phase. The genset stators are then coupled to the main 34.5kV buses through GCBs, GSUs, and ICBs. The GCBs are the synchronizing breakers, and the Gensets’ 480V AUX XFMRs are down stream of the GCBs – This allows the plant Operator to close the ICB and energize the gensets’ AUX systems so long as the main bus is energized.

The Main 34.5kV Bus is also coupled to Site AUX XFMR ICBs, an ABB AMS 1400 Synchronous Condenser ICB, and the customer FCBs. The site AUX XFMRs supply power to the Site LV E-House which distributes power back to the Gas Yard and Urea Yard as well as all other site-level AUX loads (Sync Con E-House, Ops Building, WIC, Site Lighting, Etc.).

The load side of the Customer FCBs is the point of interconnection with the customer’s MV electrical distribution system.

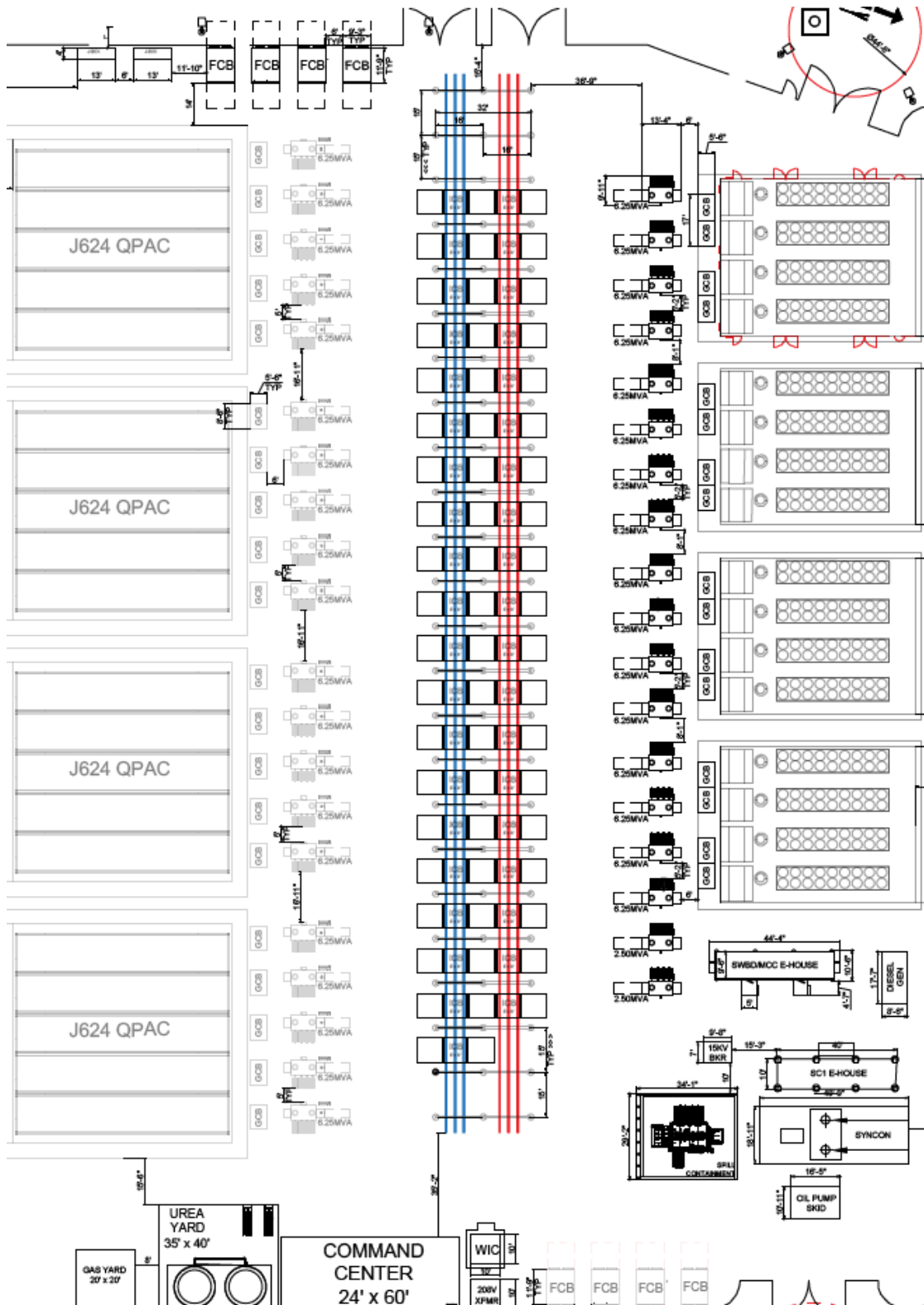


Figure 3: Typical VoltaGrid Data Center Power Plant Site Layout

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4. REFERENCES

VoltaGrid Data Center Power Plant Control Systems should be designed with reference to the following:

- VoltaGrid Site-Specific Site Layouts
- VoltaGrid Site-Specific IP Address Maps
- VoltaGrid Site-Specific MV SLDs
- VoltaGrid Site-Specific LV SLDs
- VoltaGrid Site-Specific Fuel Gas Distribution PFDs
- VoltaGrid Site-Specific Bulk Urea Distribution PFDs
- VoltaGrid Site-Specific OT Network and Control Topologies
- VoltaGrid Site-Specific IFCs
- VoltaGrid Message Bus Alarm/Alert List
- INNIO Jenbacher J620 Genset Package Technical Documentation
- INNIO Jenbacher J624 Genset Package Technical Documentation
- HUG SCR Technical Documentation
- HUG Bulk Urea Distribution System Technical Documentation
- Algas Gas Distribution Skid Technical Documentation
- Caterpillar C32B Diesel Genset Package Technical Documentation
- SEL Electrical Protection Relay Technical Documentation
- SEL RTAC Technical Documentation
- NSA Data Center Power Plant 480V E-House Technical Documentation
- ABB AMS 1400 Synchronous Condenser Technical Documentation
- ABB AMS 1400 580V E-House Technical Documentation
- REV GCB/ICB/FCB Technical Documentation
- OEM/IED-specific Modbus Register Maps and other soft I/O interface documentation
- Ethernet-based communication protocol standards, specifications, and other documentation
- NFPA 70 - National Electrical Code
- NFPA 497 - Recommended Practice for the Classification of Flammable Liquids, Gases, or Vapors and of Hazardous (Classified) Locations for Electrical Installations in Chemical Process Areas
- IEEE 1547 - Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces

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5. DEFINITIONS AND ABBREVIATIONS

The following definitions are applicable to this document:

- **Alternating Current (AC)** – A type of electrical current where the direction of electron flow reverses back and forth at regular intervals, typically creating a sine wave pattern.
- **Alternator** – A machine including a rotor, stator, and excitation system which receives mechanical energy from an Engine, converts it to electrical energy, and transmits it to the main bus via terminations, bussing, cables, breakers, XFMRs, and SWGR.
- **Application Programming Interface (API)** – A set of rules and protocols that allows two different software programs to communicate with each other and exchange data.
- **Auxiliary (AUX)** – Secondary, supplementary, complementary, or additional.
- **Automatic Transfer Switch (ATS)** – An intelligent electrical device that automatically detects a power outage from the main power source and instantly switches electrical load to a backup power source ensuring seamless, uninterrupted power, then switches back when the primary power source returns, all without human intervention.
- **Automatic Voltage Regulator (AVR)** – An electronic device that maintains a steady, constant voltage level for electrical equipment, regardless of fluctuations in the input power source, by automatically sensing changes and adjusting the output to prevent damaging sags, surges, or spikes, ensuring reliable power for sensitive electronics, generators, and industrial systems.
- **Balance of Plant (BOP)** – All AUX components, systems, and infrastructure essential for a power plant or industrial facility to operate, excluding the primary energy conversion equipment (like generators or turbines).
- **Balance of Plant Management System (BOPMS)** – The portion of the Control Room HMI which allows the Operator to monitor the status of all BOP equipment and distribution telemetry. If a single redundant BOP device has failed, the Operator should be able to see the revised system flows/states from the BOPMS.

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- **Battery Energy Storage Systems (BESS)** – A type of DER consisting of batteries, inverters, and control systems intended to provide peak-shaving, black-start power, bridge power, minimum Genset load management, and/or grid stability functions.
- **Communications and Control Management System (CCMS)** – The portion of the Control Room HMI which allows the Operator to monitor the status of all communications (OT Network) and Control (SEL 3555, 3350, 2411, 2240, 700G, 787, 751, Genset Controllers, VoltaGrid Cluster, Sync Con CCP, etc.) equipment. If a single redundant communications or control device has failed, the Operator should be able to see the revised communications flow from the CCMS.
- **Continuous Waveform Monitoring (CWM)** – A high-resolution, gapless, and real-time data streaming technology from SEL that captures voltage, current, and energy data, overcoming the limitations of trigger-based systems. CWM allows SWM to stream data at resolutions at and above 3 kHz.
- **Control Room HMI** – The application which organizes and displays Plant Control System data for the operator. The Control Room HMI also provides the operator with some system control inputs (Open/Close breaker, start/stop generator, etc.) and retransmits operator control inputs to the Master Controller. The Control Room HMI is hosted by the Cluster and accessible locally via the local OT Network and remotely via the Gateway Server.
- **Controller** – An IED which is intended to monitor telemetric devices and send control signals to electromechanical control devices and other Intelligent Electronic Control Devices.
- **Current Transformer (CT)** – An instrument transformer that "steps down" high AC currents in power systems to a lower, safer, and standardized level (like 5A or 1A) for measurement by meters, relays, and monitoring devices.
- **Distributed Energy Resource (DER)** – Any machine, not including Gensets, which is connected via an alternator or inverter to the main bus. These include but are not limited to generators, synchronous condensers, BESS, Ultracapacitors, STATCOMs, etc.
- **Direct Current (DC)** – The unidirectional flow of electric charge, meaning the current moves consistently in one direction, typically from a negative to a positive pole. Voltage is typically static in DC systems.

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- **Distributed Network Protocol (DNP)** – An open-standard communication protocol primarily used in industrial control systems to enable reliable, standardized data exchange between master controllers and IEDs.
- **Electromechanical Control Device** – Any piece of equipment that receives a control signal from an IED and performs an action that changes the behavior, state, or performance of machinery. These include but are not limited to electromechanically operated valves, governors, breakers, exciters, contactors, VFDs, etc.
- **Emergency Shut Down Valve (ESDV)** – An automated, high-integrity, valve designed to stop the flow of hazardous materials in critical situations like fires, leaks, or system failures.
- **Engine** – A Machine which converts fuel to mechanical rotation and may be run at constant speed and coupled to an Alternator for conversion to electrical energy.
- **Engineering, Procurement, and Construction Company (EPC)** – A company providing engineering, procurement, and construction services.
- **Factory Acceptance Test (FAT)** – A crucial quality assurance step where new industrial equipment or systems are tested at the manufacturer's site to ensure they meet all agreed-upon specifications and design requirements before being shipped to the customer.
- **Feeder Circuit Breaker (FCB)** – A Breaker connected between the Main Bus and the point of interconnection with the Customer's Electrical Distribution System.
- **Firewall** – Inspects network packets (small chunks of data) and allows/denies data traveling between devices or networks.
- **Front End Processor (FEP)** – Any PLC(s), PC(s), and/or application(s) which receives data from IEDs and organizes, reformats, and/or retransmits it for consumption by another device or application. In the typical VoltaGrid Data Center Power Plant FEPs are used to convert NGVL and IEC 61850 messages between the Master Controller and SEL 700G/SEL 2411.

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- **Generator Circuit Breaker (GCB)** – A breaker which connects its associated Genset Package to the main bus, or to a GSU.
- **Generator Step-Up Transformer (GSU)** – A Delta-Wye Step-Up Transformer which raises the voltage from the GCB to Main Bus voltage before connection to an ICB.
- **Genset** – The combined Engine and alternator and their associated telemetry, electromechanical control devices, and Controllers.
- **Genset Package** – The Genset, Genset enclosure, and all internal-associated AUX systems and controls. May or may not include an SCR.
- **Generic Object-Oriented Substation Event (GOOSE)** – A high-speed, reliable communication protocol/standard within the IEC 61850 framework for electrical substations, enabling IEDs to exchange critical status, alarms, and control signals instantly using Ethernet for faster automation and protection.
- **Hardware-In-The-Loop (HIL)** – A technique for verifying embedded software and controllers by connecting them to a real-time simulation of the physical environment, such as a "plant" or machinery.
- **Human Machine Interface (HMI)** – An application which organizes and displays equipment data to the operator. An HMI may also provide the operator with some system control inputs (Open/Close breaker, start/stop generator, etc.) and retransmit operator control inputs to appropriate IEDs.
- **Incremental Reserve Margin (IRM)** – The additional power output a generator or system can provide within a very short timeframe, typically 5 to 10 seconds, often critical for islanded, Micro-Grid, power generation systems.
- **Institute of Electrical and Electronics Engineers (IEEE)** – the world's largest technical professional organization, advancing technology for humanity through research, publications, conferences, and crucial industry standards (like Wi-Fi and Ethernet) in electrical, computer, and electronics engineering.

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- **Intelligent Electronic Device (IED)** – Any PLC or PC intended for the supervision and/or automatic control of a specific machine or plant subsystem. These include but are not limited to Electrical Protection Relays, Genset Controllers, Engine Controllers, SCR Controllers, Gas Distribution Controllers, Bulk Urea Distribution System Controllers, etc.
- **International Electrotechnical Commission (IEC)** – the world's leading organization that prepares and publishes international standards for all electrical, electronic, and related technologies.
- **Inter-Range Instrumentation Group – Code B (IRIG-B)** – A traditional, physical-layer time synchronization method requiring dedicated coaxial or twisted-pair cabling. It offers 1 μ s to 1ms accuracy (depending on modulation) and remains popular in legacy utility, military, and aerospace applications.
- **Isolation Circuit Breaker (ICB)** – A Breaker which provides differential protection to a GSU and connects it to the Main Bus.
- **Issued For Construction Drawing (IFC)** – A drawing/package that has been finalized and formally issued so contractor(s) may build from it.
- **Low Voltage (LV)** – Less than 1,000 V.
- **Lower Explosive Limit (LEL)** – The minimum concentration of a flammable gas or vapor in the air, expressed as a percentage, that will ignite and support combustion when exposed to an ignition source.
- **Main Bus** – The bus which feeds the customer feeder breakers and is typically rated at 34.5kV.
- **Manual Control Device** – Any control device intended to be physically manipulated by an operator which may send control signals to an IED, directly to an electromechanical control device, or in some cases, such as hand-operated valves, they may also directly manipulate machinery without the need for an electronic signal. These include but are not limited to breaker remote operation switches, breaker racking mechanisms, Engine start/stop buttons, breaker open/close buttons, hand-operated valves, etc.

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- **Manufacturing Message Specification (MMS)** – A standardized client/server communication protocol within the IEC 61850 framework used to exchange data between IEDs and higher-level systems like HMIs.
- **Master Controller(s)** – The pair of redundant controllers which automatically supervise the system and perform system-level control algorithms (Load Sharing, black start sequences, generator load management, etc.) without the need for operator intervention.
- **Medium Voltage (MV)** – Greater than or equal to 1,000 V and less than or equal to 69,000 V
- **Message Queuing Telemetry Transport (MQTT)** – A lightweight, open-standard publish-subscribe messaging protocol designed for low-power, low-bandwidth, and unreliable networks.
- **Motor Control Center (MCC)** – A centralized, modular assembly of one or more enclosed vertical sections containing motor starters, circuit breakers, fuses, and associated control equipment.
- **National Electrical Code (NEC)** – The U.S. standard for safe electrical wiring and installations.
- **National Fire Protection Association (NFPA)** – A global non-profit that creates and maintains over 300 consensus codes and standards for preventing death, injury, and economic loss from fire, electrical, and related hazards, serving as a benchmark for safety in building codes, electrical systems (like the NEC), and fire prevention strategies worldwide.
- **Network Global Variable List (NGVL)** – A high-speed Ethernet communication protocol developed by SEL to share real-time tag data between controllers.
- **Network Time Protocol (NTP)** – Time Synchronization Protocol which is widely used for standard IT applications, providing millisecond-level accuracy (usually 1-10ms). It runs over standard Ethernet networks and is highly flexible, although less precise than PTP.
- **Open Source Ethernet Protocol** – A non-proprietary, industry standard, ethernet communications protocol with a well-documented, fully open, standard/specification allowing for integration with a third party servers, PLCs, and applications. Examples include MQTT, Modbus, IEEE C37.118, IEC 61850, DNP3, etc.

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- **Operational Technology (OT) Network** – The communication network(s) which support communications between the IEDs, the Master Controller(s), and the VoltaGrid Cluster.
- **Original Equipment Manufacturer (OEM)** – A company that produces parts, components, or equipment used in another company's final, branded product.
- **Personal Computer (PC)** – A multi-purpose computer designed for individual use in homes, offices, and industrial facilities.
- **Plant Control System** – The system comprised of telemetric devices, electromechanical control devices, manual control devices, IEDs, OT Networks, Master Controllers, HMIs, Time-Series Trending Applications, Data Aggregators, Gateway Servers, Historians, and databases which allows for the supervision, automatic control, remote control, and manual control of the Data Center Power Plant and all of its AUX systems.
- **Potential Transformer (PT)** – An instrument transformer that steps down high system voltages to a safe, standardized, low level for measuring, metering, and protecting equipment in power systems.
- **Power Management System (PMS)** – A digital, networked solution that monitors, controls, and optimizes electrical power generation and distribution within a facility.
- **Process Flow Diagram (PFD)** – A high-level engineering drawing that shows the major pieces of equipment in a process and how material and energy flow between them. It's used to communicate the overall process concept and mass/energy routing without the construction-level detail found on a P&ID.
- **Programmable Logic Controller (PLC)** – A ruggedized, industrial-grade computer designed to automate manufacturing processes and machinery.
- **Precision Time Protocol (PTP)** – Time Synchronization Protocol which provides sub-microsecond or nanosecond-level accuracy, making it ideal for high-precision industrial automation, financial trading, and modern power utility substations. It uses hardware timestamping and requires PTP-capable network switches.

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- **Real Time Automation Controller (RTAC)** – A purpose-built automation controller used in power systems that implements real-time supervisory and integration logic (protocol translation, permissives/interlocks, data aggregation, and control sequencing) and is commonly used as a front-end processor or master controller for substations and power plants.
- **Real Time Digital Simulator (RTDS)** – A specialized, high-performance computing platform used to simulate power systems in real-time for testing and analysis.
- **Schweitzer Engineering Laboratories (SEL)** – A global, employee-owned, company that designs and manufactures digital products and systems for protecting, automating, monitoring, and controlling electric power systems.
- **Selective Catalytic Reduction (SCR) Package** – an assemblage of ducting, chambers, telemetric devices, and controls which utilize catalysts and reductants to minimize exhaust gas emissions.
- **Simple Network Management Protocol (SNMP)** – A lightweight, widely-used protocol for monitoring and managing devices on an IP network. It lets a central manager poll devices (agents) for status/measurements, receive asynchronous alerts, and (where allowed) change configuration values.
- **Single Line Diagram (SLD)** – A schematic electrical drawing that shows an electrical power system with one simplified line per three-phase circuit. It's the canonical, high-level electrical diagram used to show how generators, transformers, busbars, breakers, meters and feeders are connected and protected – together with ratings and tag/identifiers – without drawing every physical conductor or mechanical detail.
- **Site Acceptance Test (SAT)** – A final, system-level, check performed after construction to verify a newly installed system or equipment works correctly in its intended environment.
- **Static Synchronous Compensator (STATCOM)** – A fast-acting power electronics device that regulates voltage and stabilizes power grids by dynamically injecting or absorbing reactive power, acting as a controllable source or sink for reactive power to maintain grid voltage, improve power factor, and enhance transmission capacity.

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- **Switchboard (SWBD)** – A LV central electrical panel that receives incoming power and distributes it safely to various circuits within a building or facility, featuring switches, circuit breakers, fuses, and busbars for control, protection, and organization of electricity flow.
- **Switchgear (SWGR)** – a MV assembly, completely enclosed on all sides and top with sheet metal (except for ventilating openings and inspection windows), and containing primary power circuit switching, interrupting devices, or both, with buses and connections. It is designed for controlling, metering, and protecting electrical equipment.
- **Synchronous Condenser (Sync Con)** – A rotating electrical machine, like a generator but not connected to a prime mover (turbine), used to stabilize the power grid by controlling voltage, providing inertia, and managing reactive power to balance supply and demand, especially as intermittent renewable energy sources grow.
- **Synchrowave Monitoring (SWM)** – a software solution from SEL that provides high-resolution, gapless, and continuous recording and visualization of power system data, including voltage, current, and frequency.
- **Telemetric Device** – Any sensor or contact intended to provide digital or analog statuses and/or measurements to an IED. These include but are not limited to pressure switches, level switches, pressure sensors, temperature sensors, PTs, CTs, speed sensors, status contacts, flow meters, level sensors, etc.
- **Time-Series Trending Application** – Any application which uses a data historian to display time-series data to improve operator situational awareness, assist troubleshooting efforts, and expose system level performance criteria. SWM is a typical Time-Series Trending application used by VoltaGrid. SWM is typically hosted by the Cluster and accessible locally via the local OT Network and remotely via the Gateway Server.
- **Transformer (XFMR)** – A static electrical device that transfers AC electrical energy between circuits through electromagnetic induction, increasing (step-up) or decreasing (step-down) voltage levels without altering frequency
- **Variable Frequency Drive (VFD)** – An electronic motor controller that drives an AC motor by varying the input frequency and voltage to precisely control speed and torque.

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- **Virtual Domain (VDOM)** – A virtualization feature commonly found in next-generation firewalls (e.g., Fortinet FortiGate appliances) or similar multi-tenant security devices. They partition a single physical appliance into multiple independent "virtual firewalls," each with its own configuration, policies, routing tables, and administrative access.
- **Virtual Local Area Network (VLAN)** – operates at Layer 2 (Data Link layer) of the OSI model and is typically implemented on Ethernet switches. VLANs allow a single physical switch to be divided into multiple isolated broadcast domains, effectively creating "virtual" networks without needing separate physical cabling.
- **Volt-Amps (VA)** – Unit of measurement for “apparent” electric power. Can be calculated by dividing the measured real power by the power factor.
- **Volt-Amps-Reactive (VAR)** – Unit of measurement for “reactive” electric power.
- **VoltaGrid Cluster** – The Kubernetes Cluster of On-Site servers that host VoltaGrid’s Control Room HMI, VoltaGrid Message Bus, VoltaGrid Gateway Server, Historians, and Time-Series Trending application(s).
- **VoltaGrid Gateway Server** – The application which receives and/or formats system data for transmittal to cloud storage. The VoltaGrid Gateway Server is hosted by the VoltaGrid Cluster. The VoltaGrid Gateway Server also allows VoltaGrid employees to remotely access the VoltaGrid Cluster, Control Room HMI, Time-Series Trending applications, and select IEDs remotely.
- **VoltaGrid Message Bus** – The VoltaGrid specific Data Aggregator which includes an MQTT Broker as well as several other applications which receive Modbus data from all Plant IEDs and converts it to MQTT data (in the VoltaGrid Message Bus) so that it can be accessed/used by the Control Room HMI and VoltaGrid Gateway Server. The Message Bus is hosted by the VoltaGrid Cluster and accessible locally via the local OT Network and remotely via the Gateway Server.
- **Walk-In-Closet (WIC)** – In the context of VoltaGrid Data Center Power Plants, this is the building which houses the VoltaGrid Cluster and the power plant’s central network equipment including but not limited to switches, firewall, and time synchronization equipment.
- **Watts (W)** – Unit of measurement for “real” electric power.

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6. PLANT CONTROL SYSTEM OBJECTIVES

The following sections define the objectives that the here-defined Plant Control System requirements are intended to achieve.

6.1. GENERATOR AND DER START/STOP

The Plant Control System shall provide means to start and stop generators/DERs locally, remotely (via the Control Room HMI), and automatically depending on site/plant conditions – To including start/stop control for the backup diesel generator for emergency AUX power.

6.2. GENERATOR AND DER LOCAL AND REMOTE MODE SELECTION

The Plant Control System shall provide the Operator with the ability to block start and control the operating mode (manual, auto, local, remote) for each generator/DER both from the local HMI/control panel and from the Control Room HMI.

6.3. GENERATOR AND DER SYNCHRONIZATION

If the primary or secondary common bus is energized prior to an Engine start sequence and the machine's ICB is closed, the Plant Control System shall automatically synchronize the machine to the common bus and then close the GCB. For gensets, this function should be performed by the Genset Controller.

If the machine's ICB is open, and the Main bus is energized, the control system should block close on the associated Genset GCB.

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6.4. FREQUENCY/VOLTAGE REGULATION

The Plant Control System shall attempt to maintain generator, common bus, and customer feeder breaker voltages and frequencies at nominal values (13.8kV or 34.5kV and 60Hz). During and after transient events, the control system shall endeavor to restore nominal voltages and frequencies as quickly as possible.

VoltaGrid's frequency/voltage stability targets are as follows:

- Transient frequency dips at the Main Bus shall be no less than 58.2 Hz (97% of nominal)
- Transient Frequency swells at the Main Bus shall be no greater than 61.2 Hz (102% of nominal)
- Transient Phase-Phase Voltage dips at the Main Bus shall be no less than 32.8kV (95% of nominal)
- Transient Phase-Phase Voltage swells at the Main Bus shall be no greater than 36.2kV (105% of nominal)

VoltaGrid will design/select/add DERs in the MV system to ensure the above is physically possible, however, the Master Controller, Genset Controllers, and DER Controllers shall be capable of dispatching these resources as needed to meet the requirement.

6.5. GENERATOR AND DER LOAD SHARING

The Plant Control System shall utilize speed and voltage bias signals generated by the Master Control System to balance the real and reactive power loads across all generators and to utilize DERs to minimize voltage/frequency disturbances as much as possible.

6.6. MEDIUM VOLTAGE BREAKER OPEN/CLOSE

The Plant Control System shall provide means to open and close all MV breakers and disconnect switches locally, remotely, and automatically depending on site/plant conditions.

Note: The operator should not be able to initiate a GCB closure except by initiating an Engine start sequence, after which the Genset Controller and generator protection relay will automatically synchronize the Genset to the main bus and close the GCB.

The Plant Control System shall provide the Master Controller and Operator (via the local HMI or Control Room HMI) with the ability to block close for each breaker independently.

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6.7. MANUAL BACKUP CONTROL

In General, the Plant Control System should include sufficient Manual Backup Control mechanisms such that the Operator can still control the plant in the event of a complete Control Room shutdown including a loss of power to both Master Controllers and the VoltaGrid Cluster.

For all failures which do not present a safety issue for personnel, or risk of equipment damage, and that do prevent the operator from starting/stopping generators or opening/closing breakers through the normal HMI interfaces; The Plant Control System shall provide a backup, manual, means of generator start/stop and breaker open/close control.

For generators, there should be local run/stop buttons and mode selector switches on the local Genset control panel or local Genset Controller HMI. For breakers, excluding GCBs, the Plant Control System shall include remote operation switches (Chicken Switches) which may be manually operated to open and close the breaker (provided that a block close is not asserted).

6.8. ELECTRICAL PROTECTION

The Plant Control System shall make every attempt possible to prevent injury to personnel, damage to equipment, and system -wide blackouts during electrical failures. Electrical protection relay settings should be developed in accordance with all applicable equipment data sheets/ratings, Arc Flash and Coordination Studies, customer requirements, and standard industry practices.

Note: GOOSE-based GSU Differential Protection and Breaker Failure Protection shall be provided by the GCB and ICB Electrical Protection Relays.

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6.9. EQUIPMENT PROTECTION

The Plant Control System should make every attempt possible to prevent injury to personnel, damage to equipment, and system -wide blackouts during mechanical failures. Alarm levels, emergency shutoffs, and Engine shut-down circuits should be set in accordance with all equipment data sheets/ratings, HAZOP results, customer requirements, and standard industry practices.

6.10. AUTOMATIC BLACKOUT RECOVERY

The Plant Control System should automatically execute the necessary control sequences required to ensure that critical AUX/LV loads are immediately reenergized after a system-wide blackout.

The status of the Blackout Recovery control sequence shall be provided to the operator via the Control Room HMI.

6.11. AUTOMATIC BLACK START

The Plant Control System shall support an automatic black start control input (via the Control Room HMI) which will initiate an automated Black Start sequence executed by the Master Controller. When the Black Start sequence is initiated, the Master Controller should systematically start/synchronize generators and open/close SWGR in order to restore the plant to normal operations after a black out event.

The status of the Black Start control sequence shall be provided to the operator via the Control Room HMI.

6.12. LOAD SHEDDING

If required, the Plant Control System should be capable of Load Shedding (opening specific breakers to maintain frequency stability) on its own, or in response to an Open Source Ethernet Protocol message from the customer's electrical distribution control system.

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6.13. GENSET PRIORITY MANAGEMENT

The Plant Control System shall provide the Operator with the means to assign a priority level to each genset in the plant (via the Control Room HMI). There should be 100 priority levels – ensuring the Operator can assign a different priority level to every Genset if desired. The Operator should also be able to assign the same priority level to multiple Gensets if desired.

These priority levels will determine which Genset the Master Controller will automatically start/stop for IRM and Minimum Genset Load Management.

6.14. INSTANTANEOUS RESERVE MARGIN MANAGEMENT

The control system shall automatically start, synchronize, and close-in additional Gensets to the common bus (so long as at least one Genset is stopped and set to start automatically) as required to maintain an instantaneous reserve margin that satisfies the customer needs according to anticipated load profiles.

When a generator start is initiated to satisfy IRM requirements, two generators should be started and synchronized to the common bus. If both generators succeed in their start sequences and synchronize to the common bus, then the generator with the lowest priority should be unloaded and shut down.

If neither generator succeeds in its start sequence and synchronization, then (if available) two more generators should attempt to start and synchronize to the common bus.

All pertinent IRM Management statuses shall be provided to the operator via the Control Room HMI.

6.15. MINIMUM GENSET LOAD MANAGEMENT

The control system shall automatically unload generators and open their GCBs as necessary in order to maintain a minimum load on all running/synchronized generators. If IRM and minimum loading requirements are in conflict, IRM requirements shall take priority, and no generators shall be unloaded/disconnected from the common bus.

All pertinent Minimum Load Management statuses shall be provided to the operator via the Control Room HMI.

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6.16. PARTY MODE

The Plant Control System shall provide Operator with the means to place the plant in “Party Mode” – starting all generators and DERs (except those which are Blocked locally or remotely) and bypassing the IRM and Minimum Load Management functions.

Party Mode should be actuated via a button on the Control Room HMI and executed by the Master Controller.

6.17. FUEL GAS DISTRIBUTION

The Plant Control System shall control the Fuel Gas Distribution System as needed to maintain Genset Package inlet gas valve pressures and temperatures as needed for uninterrupted Power Plant operations.

Emergency Stop Pushbuttons (strategically located throughout the plant) shall have series dry contacts in the Fuel Gas Distribution System’s ESDV Control Loop.

All pertinent Fuel Gas Distribution system data (valve positions, pressures, temperatures, etc.) shall be provided to the operator via the Control Room HMI.

6.18. EMISSIONS MANAGEMENT

The Plant Control System shall control the Bulk Urea Distribution System and SCRs as needed to maintain contractual emissions obligations.

Bulk Urea Distribution System control should be provided by the Bulk Urea Distribution System Controller and should not require control input from the Master Controller.

SCR system control shall be provided by the Genset Controllers or by an independent SCR Controller. In either case, SCR control should not require control input from the Master Controller.

All pertinent Emissions and Bulk Urea Distribution System data (tank levels, dosing rates, valve positions, pressures, temperatures, etc.) shall be provided to the operator via the Control Room HMI.

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6.19. LOW VOLTAGE POWER DISTRIBUTION MONITORING AND CONTROL

The Plant Control System shall monitor and control LV equipment as necessary to support Operator Situational Awareness, remote control requirements, and Blackout Recovery control sequences.

All pertinent LV Distribution system data (breaker positions, voltages, frequencies, currents, etc.) shall be provided to the operator via the Control Room HMI.

6.20. CONTROL POWER MONITORING

The Plant Control System shall monitor the status of all control power circuits (DC and AC) and their associated battery chargers, batteries, and/or UPS.

The output voltage and status of all control power battery chargers, batteries, and UPS shall be provided to the operator via the Control Room HMI.

6.21. J620 MOBILE GENSET INTEGRATION

The Plant Control System shall provide the means to integrate multiple VoltaGrid Mobile J620 Genset Trailers into the Fuel Gas Distribution and MV Distribution Systems.

All pertinent mobile J620 Genset data (GCB Status, Running/stopped status, etc.) shall be provided to the operator via the Control Room HMI.

6.22. DATA CENTER TELEMETRY/FEEDBACK CONTROL MONITORING

The Master Controller shall be capable of monitoring and responding to Forward Looking Feedback signals from the Customer/Data Center.

The signals and control responses are to be defined by VoltaGrid. The status of these signals shall be provided to the Operator by the Control Room HMI.

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6.23. DATA TRANSFER, ACQUISITION, AND STORAGE

The Plant Control System, and all of its constituent IEDs, shall expose all pertinent data (signal values, equipment states, control output values, internal process variables, alarms, etc.) to the OT Network via an Open Source Ethernet Protocol.

The IEDs, Master Controller(s), and VoltaGrid Cluster shall have access to any/all of this data, at all times, as needed to achieve the Plant Control System objectives defined in this document.

The VoltaGrid Message Bus shall aggregate all plant

The IEDs, Master Controller(s), and VoltaGrid Cluster shall be time synchronized to an IRIG-B time source or time synchronized to a central clock to maintain time alignment amongst data points and alarms from all devices, systems, and subsystems.

The OT Network shall be designed to the following specifications (at a minimum) to ensure seamless, continuous, low-latency control and monitoring of data transfer between devices and applications:

- 1GB/s Throughput
- 100GB/s Routing Throughput
- Less than 10ms Latency
- Less than 4ms Latency for GOOSE communications

The VoltaGrid Gateway Server shall transmit all Plant Control System data to the VoltaGrid cloud-based database for long term storage and view portal population. The VoltaGrid Cluster shall also support a 30 Historian for SWM.

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6.24. HUMAN MACHINE INTERFACES

The control system shall include HMIs which give the operator visibility of critical, real-time, equipment data/statuses as well as limited control inputs.

In general, the operator should be able to initiate/control the following items from an HMI:

- Open/Close breakers (except for GCBs) provided that a block close is not asserted by the Master Controller or electrical protection relay
- Set specific Gensets to “Run”, “Stop”, “Auto”, and “Block”
- Initiate black start and bus transfer sequences

A centralized Control Room HMI should provide all data, alarms, and control functions to the Operator (from all Plant IEDs) that are required to control the plant during normal operations and maintain Operator situational awareness.

6.25. TIME-SERIES TRENDING

The Plant Control System shall include at least one Time-Series Trending Application to improve Operator situational awareness. This objective shall be satisfied via an instance of SWM hosted by the VoltaGrid Cluster as well as by the Control Room HMI.

The Time-Series Trending Application(s) shall have access to all pertinent data via Open Source Ethernet Protocols.

6.26. CONTINUOUS WAVEFORM MONITORING

The Plant Control System shall support CWM at a 3kHz sample rate from SEL 735 Power Meters to allow for in depth analysis of Main Bus harmonics up to the 25th order.

This data should also allow for in-depth analysis of sub-transient voltage and frequency deviations during rapid load changes and switching operations.

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6.27. RELIABILITY

In General, every effort should be made to prevent a single point of failure. Single device failures shall not result in system-wide blackouts, and efforts shall be made to duplicate signals where necessary to prevent reduced remote control capabilities and/or loss of data in HMI(s), Time-Series Trending Application(s), and Gateway Server(s).

Wherever possible, single device failures should only result in an alarm. When possible, a failure of two (both) redundant devices should result in a loss of visibility, limp-mode operations, or a single unit out of N+2 shutdown but not a system-wide blackout.

For mechanical device failures which do not present a safety concern, the device should fail in an operational state (breaker closed/valve open).

For mechanical device failures which do present a safety concern, the device should fail into a non-operational/safe state (breaker open/valve closed).

The Plant Control System shall not infringe upon the Power Plant's ability to maintain a 99.9% up-time.

Any required maintenance items shall allow for concurrent serviceability. Maintenance items shall not require a system-wide blackout.

6.28. ALARM/ALERT MANAGEMENT

All Alarms and Alerts, from all Plant IEDs, should be collected and organized into a uniform format by the VoltaGrid Cluster/Message Bus.

6.29. RTDS SIMULATION AND HIL FAT VERIFICATION

System stability requirements and Master Controller functional requirements should be verified by the Master Controller OEM via RTDS simulations and HIL FAT Tests prior to the Plant design finalization.

Further detail is provided in the [Test Reports](#) section.

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7. MEDIUM VOLTAGE GENSETS

The MV Gensets in a typical VoltaGrid Data Center Power Plant are the primary means of power generation. All other assets, equipment, and AUX/BOP systems are in place to ensure, support, or otherwise assist with the MV Gensets' ability to support the customer's requirements (load, IRM, Emissions, main bus frequency, main bus voltage, etc.)

The Plant Control System should also make every effort to ensure, support, or otherwise assist with continuous, stable, operation of the MV Gensets – It should not impede the continuous operation of the MV Gensets in any way.

The typical MV Gensets used in VoltaGrid Data Center Power Plant designs are the INNIO Jenbacher J620 and J624 Gensets which includes a DIANE Local Control Panel and Genset Controller. In addition to local, manual control interfaces, DIANE hosts a Modbus Server which can be used for remote monitoring.

The DIANE customer control interface is primarily hard-wired. Some control related feedback signals are exposed to the Master Controller and Control Room HMI via the DIANE Modbus Server.

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7.1. TELEMETRIC DEVICES

MV Genset Packages shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Pressure switches/sensors
- Temperature switches/sensors
- Vibration sensors
- Phase CTs
- Phase PTs
- Speed sensor(s)
- Level switches/sensors
- Status Contacts (I.e. running/stopped)

The number and type of these devices and the signals provided by them shall be defined by the Genset OEM, and all Genset telemetric device signals shall be connected to the Genset Controllers or Genset sub-system Controllers.

7.2. ELECTROMECHANICAL CONTROL DEVICES

In addition to Starters, Governors, and Exciters - Genset Packages shall include a plurality of other electromechanical control devices required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Contactors
- VFDs
- Control Valves
- Electronically Controlled Clutches

The number and type of these devices and their signal requirements shall be defined by the Genset OEM, and all Genset control signals shall be provided by the main Genset Controller or Genset sub-system Controller.

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7.2.1. Engine Starters

Engine Starter Motors shall be remotely actuated by the Engine and/or Genset Controllers when requested by the Operator (via the Control Room HMI or a Manual Control Device) or the Master Controller provided that the Genset is in AUTO and no Block Start is condition is present.

The Engine Starter Motor control signal type shall be defined by the Engine OEM. Engine and/or Genset Controller start/stop signal values shall be exposed to the OT network via an Open Source Ethernet Protocol.

7.2.2. Engine Governors

Engine Governors are critical electromechanical control devices in VoltaGrid Data Center Power Plants as they are the primary means of controlling Genset real power output levels and maintaining the system frequency. The Genset governors shall receive throttle position/speed control signals from the Engine Controller.

The Engine/Genset Controller shall run the governor in droop by default but accept speed control biases from the Master Controller (or from an IED retransmitting Master Controller control signals) to maintain the Main Bus at 60Hz for all load levels. During normal operations the Master Controller will run the Gensets in a communications-assisted Droop-Sharing mode for real power.

The Governor control signal type/range shall be defined by the Engine OEM. Engine and/or Genset Controller governor control signal values shall be exposed to the OT network via an Open Source Ethernet Protocol.

7.2.3. Alternator Exciters

Alternator Exciters are also critical electromechanical control devices in VoltaGrid Data Center Power Plants as they are the primary means of controlling Genset reactive power output levels and maintaining the system voltage. The alternator excitation control systems receive voltage control signals either directly from the Genset Controller or indirectly from the Genset Controller via an intermediary excitation Controller or AVR.

The Genset Controller or AVR shall run the Alternator in droop by default but accept voltage control biases from the Master Controller (or from an IED retransmitting Master Controller control signals) to maintain the Main Bus at 34.5kV for all load levels. During normal operations the Master Controller will run the Gensets in a communications-assisted Droop-Sharing mode for reactive power.

The excitation control system signal type/range shall be defined by the Engine OEM. Genset Controller and/or AVR excitation control signal values shall be exposed to the OT network via an Open Source Ethernet Protocol.

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7.3. MANUAL BACKUP CONTROL DEVICES

MV Gensets shall include a local genset HMI or control panel as the primary means of Manual Backup Control.

The local Genset controls shall include engine Start/Stop Buttons for manual control in the event of a Control Room HMI or Master Controller failure. These buttons may be hard-wired physical buttons or included as part of the local Genset HMI.

The local genset controls shall also include local Selectors to control the operating mode and remote accessibility for the genset. The typical buttons/selectors provided by the INNIO J620/J624 local genset control HMI (DIANE) are shown in **Figure 4**. These buttons and selectors can be accessed by pressing the “Panel” button from the local genset control HMI.

Note: These operating modes should not be confused with the operating mode selected at the Control Room HMI. A Genset’s operating mode as selected in the Control Room HMI (Block/Manual/Auto) is unrelated to its operating mode as selected from the local genset Operating panel – although it can have similar/overlapping effects. See [Control Room Human Machine Interface](#).

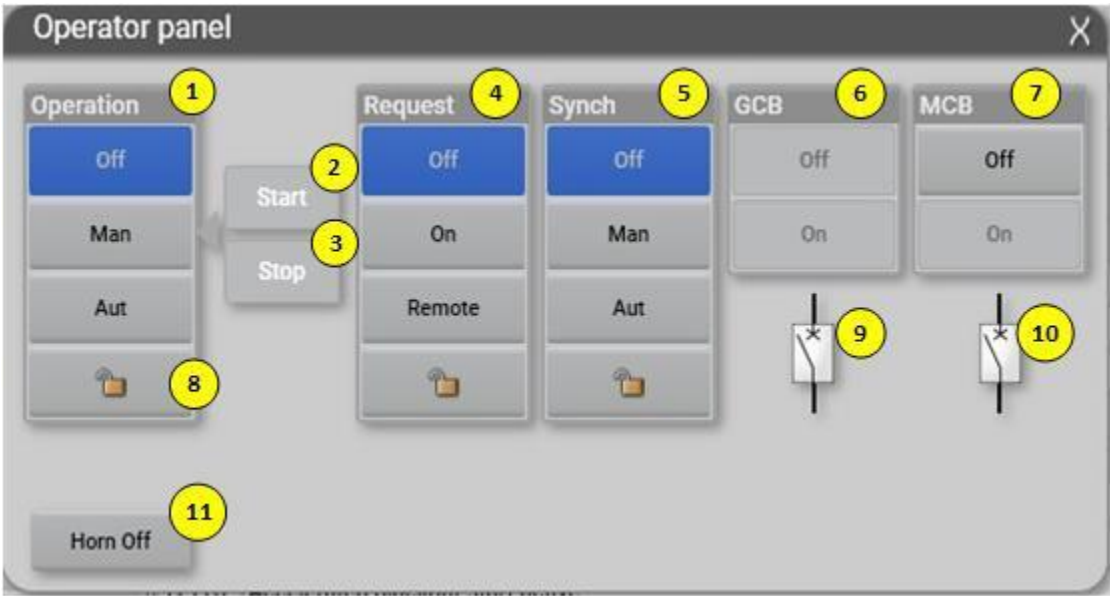


Figure 4: INNIO J620/J624 Operator Panel

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7.3.1. Operation Selector

The “**Operation**” selector buttons shown under the yellow “1” in **Figure 4** have the following functions/effects:

- **Off** – This is effectively a local Block mode. If this “Operation” mode is selected, the genset will shut down or remain shut down and all local and remote engine start signals will be disabled/ignored. The “Ready For Remote Start” Digital Output from the DIANE will remain low so long as “Operation” is set to “Off”.
- **Man** – This is effectively a local control mode. The engine may be started or stopped using the “Start” and “Stop” buttons shown next to the yellow “2” and “3” in **Figure 4**, however the “Ready For Remote Start” Digital Output from the DIANE will remain low so long as “Operation” is set to “Man”.
- **Aut** – **This is not, by itself, a remote control mode.** This selection only enables “Start” and “Stop” “Requests” as seen in the “Request” selector below the yellow “4” in **Figure 4**. Setting “Operation” to “Aut” will not cause the “Ready For Remote Start” Digital Output to go high unless the “Request” Selector is set to “Remote”.

7.3.2. Request Selector

The “**Request**” selector buttons shown under the yellow “4” in **Figure 4** have the following functions/effects:

- **Off** – If “Operation” is set to “Aut”, then setting the “Request” selector to “Off” will shut the engine down. The “Ready For Remote Start” Digital Output from the DIANE will remain low so long as “Request” is set to “Off”.
- **On** – If “Operation” is set to “Aut”, then setting the “Request” selector to “On” will start the engine. The “Ready For Remote Start” Digital Output from the DIANE will remain low so long as Request is set to On.
- **Remote** – **This is not, by itself, a remote control mode.** This selection enables start and stop “Remote Requests”, however, if the “Operation” selector is not set to “Aut” then start/stop “Requests” are not enabled. Setting “Request” to “Remote” will not cause the “Ready For Remote Start” Digital Output to go high unless the “Operation” Selector is set to “Aut”.
- **Note:** Even after the “Ready for Remote Start” signal has gone high, the remote controller (Master Controller) will still be unable to connect the genset to the Main Bus (for IRM Management purposes) unless the “Synch” selector, below the yellow “5” in **Figure 4**, is set to “Aut”. The Master Controller should not consider the Genset ready for automatic starting until the “Operator Panel” Selectors are set as follows:
 - **Operation = Aut**
 - **Request = Remote**
 - **Synch = Aut**

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7.3.3. Synch Selector

The “**Synch**” selector buttons shown under the yellow “5” in **Figure 4** have the following functions/effects:

- **Off** – The GCB is blocked and will not close. The Engine may be started but cannot synchronize to the Main Bus.
- **Man** – Synchronization can be started and stopped by pressing the “On” or “Off” buttons in the “GCB” selector shown under the yellow “6” in **Figure 4** (assuming the genset is running at rated speed and voltage).
- **Aut** – Synchronization may be remotely started/stopped (by the Master Controller).
- **Note:** As stated in the section above, the “Operator Panel” selections must be as follows to enable the Master Controller to start and synchronize Gensets for IRM Management:
 - **Operation = Aut**
 - **Request = Remote**
 - **Synch = Aut**

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7.4. INTELLIGENT ELECTRONIC DEVICES

In addition to Genset, Engine, and Excitation Controllers – MV Genset Packages shall include a plurality of other IEDs required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Contactors
- VFDs
- Control Valves
- Electronically Controlled Clutches

The number, type, and signal requirements of these devices shall be defined by the MV Genset OEM, and all Genset package control signals shall be connected to the Genset Controller or Genset sub-system controller. All control signal data points shall be exposed to the OT Network via an Open Source Ethernet Protocol.

7.4.1. Genset Controllers

In a Genset Package, the Genset Controller should work in concert with the Engine Controller and Excitation Controller to automatically monitor and control all Genset Package primary and AUX systems. These systems include but are not limited to the following:

- Engine Governor
- Alternator Excitation System
- Fuel Train
- Turbo Chargers
- Radiators
- Lube Oil Systems
- Coolant Systems
- Exhaust Systems
- SCRs

All Genset package telemetry/control signals shall be exposed to the OT Network by the Genset Controller via an Open Source Ethernet Protocol. The Genset Controller should aggregate all pertinent data from all other Genset Package IEDs. The Genset Controller shall also provide a customer control interface to send/receive feedback signals and control inputs to/from a Master Controller, Electrical Protection Relay, and/or Remote I/O Module.

Table 1 details the typical, hard-wired, customer control interface for a stationary INNIO J620 or J624 Package Genset Controller. The specific A1 Panel terminal numbers for these I/O are to be provided by INNIO. In addition to this hard-wired interface, the Genset Controller will provide some feedback signals (Engine Hours, Operator Panel selections, etc.) via a Modbus Server. INNIO will provide the associated Modbus Register Map.

Table 1: INNIO J620/J624 Hard-Wired Customer Interface

Signal	Signal Type	Notes
Power Setpoint	Analog Input	Real Power Load Sharing Control Setpoint
Voltage Setpoint	Analog Input	Reactive Power Load Sharing Control Setpoint
Fixed Power Mode	Digital Input	Constant P/Q Toggle
Active Power	Analog Output	DIANE Real Power Measurement
Reactive Power	Analog Output	DIANE Reactive Power Measurement
Available Active Power	Analog Output	Available Power (does not decrease as load increases)
Engine Demand (Start/Stop)	Digital Input	Start(high)/Stop(low) Signal
Start Permissive	Digital Input	Engine is Allowed to Start
GCB Release to Close to Dead Bus	Digital Input	Dead Bus Arbitration is Satisfied
Fast Disconnect	Digital Input	Immediate Engine Shutdown for Breaker Failure
Generator in parallel with module	Digital Input	Soft Shutdowns and Derating is Enabled
Rate Limiter Disable	Digital Input	Disables the Genset Controller P Setpoint Rate Limiter
Remote Block	Digital Input	Blocks the Genset from Starting via Local Controls
Plant Emergency Mode	Digital Input	Genset Cannot Perform a Soft Shutdown
Alarm Acknowledge	Digital Input	Acknowledges All Active Alarms
TRIP Signal	Digital Input	700G has Attempted to Trip the GCB
Overload/Short Circuit	Digital Input	700G Tripped on an Overcurrent Protection Element
Earth Fault	Digital Input	700G Tripped on a Ground Fault Protection Element
GCB Open Command	Digital Output	Connected to GCB Trip Coil
GCB Close Command	Digital Output	Connected to GCB Close Coil Through SEL 700G
Engine is Demanded	Digital Output	DIANE has received a Start Signal
Engine in Operation (running)	Digital Output	Engine is Running
GCB Opening Failure	Digital Output	DIANE Failed to Open the GCB
Ready For Remote Start	Digital Output	Genset is Ready for Remote Start Requests
Generator Trip Status	Digital Output	Engine is Stopped
Delayed Shutdown/Trip Request	Digital Output	Genset Controller is Requesting a Delayed Shutdown

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7.4.2. Engine Controllers

The Engine Controller should work in concert with the Genset Controller to automatically monitor and control all Genset Package primary and AUX systems. These systems include but are not limited to the following:

- Engine Governor
- Alternator Excitation System
- Engine Starter Motors
- Engine Fuel Train
- Turbo Chargers
- Radiators
- Lube Oil Systems
- Coolant Systems
- Exhaust Systems
- SCRs

The number and type of inputs/outputs to/from Genset Package telemetric devices, electromechanical control devices, and the Engine Controller shall be defined by the Engine OEM. All Genset Package telemetry/control signals collected by the Engine Controller shall be exposed to the Genset Controller which will expose them to the OT Network via an Open Source Ethernet Protocol.

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7.4.3. Excitation Controllers

The Excitation Controller should work in concert with the Genset Controller to automatically monitor and control all Genset Package primary and AUX systems. These systems include but are not limited to the following:

- Engine Governor
- Alternator Excitation System
- Engine Starter Motors
- Engine Fuel Train
- Turbo Chargers
- Radiators
- Lube Oil Systems
- Coolant Systems
- Exhaust Systems
- SCRs

The number and type of inputs/outputs to/from Genset Package telemetric devices, electromechanical control devices, and the Excitation Controller shall be defined by the Engine OEM. All Genset Package telemetry/control signals collected by the Excitation Controller shall be exposed to the Genset Controller which will expose them to the OT Network via an Open Source Ethernet Protocol.

7.5. FAILURE MODES

A single MV Genset failure/shutdown shall not result in a system-wide blackout. Individual Genset failures may trigger individual engine shutdowns if required.

Note: The Master Controller's IRM Management control sequence should maintain a minimum of N+2 gensets synchronized to the Main Bus – preventing an individual engine shut down from causing a system-wide blackout.

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8. DISTRIBUTED ENERGY RESOURCES

A DER is any machine, not including Gensets, which is connected via an alternator or inverter to the Main Bus. These include but are not limited to Synchronous Condensers, BESS, Ultracapacitors, STATCOMs, etc.

Synchronous Condensers are of particular interest as they are included in typical VoltaGrid Data Center Power Plant designs. The Synchronous Condenser(s) in a VoltaGrid Data Center Power Plants ensure that the Main Bus Voltage and Frequency can remain within customer tolerances during the erratic load swings created by A.I. Data Centers.

The typical Synchronous Condenser used in VoltaGrid Data Center Power Plant designs is the ABB AMS 1400 With Flywheel which includes a CCP (Local Control Panel and Sync Con Controller). In addition to local, manual, control interfaces, the CCP hosts a Modbus Server which can be used for remote monitoring and control.

8.1. TELEMETRIC DEVICES

All DER Packages shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the DER package including but not limited to the following:

- Pressure switches/sensors
- Temperature switches/sensors
- Vibration sensors
- CTs
- PTs
- Speed sensors
- Level switches/sensors
- State contacts (I.e. running/stopped)
- LEL switches/sensors
- Fire Alarm switches/sensors

The number and type of these devices and the signals provided by them shall be defined by the DER OEM, and all DER telemetric device signals shall be connected to the main DER Controller or DER sub-system Controllers.

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8.2. ELECTROMECHANICAL CONTROL DEVICES

DERs shall include a plurality of electromechanical control devices required for the safe, effective, and efficient operation and supervision of the DER and its AUX systems including but not limited to the following:

- Contactors
- VFDs
- Control Valves
- Electronically Controlled Clutches
- Exciters and/or AVR
- Inverters
- Pony Motor Drives
- Back-Up Generator Starters

The number and type of these devices and their signal requirements shall be defined by the DER OEM and all DER control signals shall be provided by the main DER Controller or DER sub-system Controllers.

8.3. MANUAL BACKUP CONTROL DEVICES

All DER packages shall include a local HMI or control panel as the primary means of Manual Backup Control.

The local DER controls shall include Start/Stop Buttons for manual control in the event of a Control Room HMI or Master Controller failure. These buttons may be hard-wired physical buttons or included as part of the local DER HMI.

The local DER controls shall also include local Selectors to control the operating mode and remote accessibility for the DER.

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8.4. INTELLIGENT ELECTRONIC DEVICES

In general, DER Controllers shall meet the same control and signal requirements as described in the [Medium Voltage Gensets](#) section of this document.

The number and type of inputs/outputs to/from DER Package telemetric devices, electromechanical control devices, and the DER Controller shall be defined by the DER OEM, and all DER package telemetry/control signals shall be exposed to the OT Network via an Open Source Ethernet Protocol.

The DER Controller shall also provide a customer control interface to send/receive feedback signals and control inputs from a Master Controller, Electrical Protection Relay, Remote I/O Module, and/or HMI.

In the case of the ABB AMS 1400 Synchronous Condenser, the CCP shall aggregate all telemetry/control data from all Sync Con AUX/sub-system controllers and expose it to the OT Network via an Open Source Ethernet Protocol. ABB will also define and provide the customer control interface.

The existing ABB AMS 1400 CCP customer control interface includes optional hard-wired and Modbus Server customer control interfaces. If the Master Control system is integrated with this customer control interface, it may utilize either the hard or soft I/O interface or both.

8.5. FAILURE MODES

A single Synchronous Condenser system failure shall not necessitate a system-wide blackout, although it may cause a system-wide blackout depending on the customer load profile.

9. MEDIUM VOLTAGE DISTRIBUTION SYSTEM

The MV Distribution System allows a typical VoltaGrid Data Center Power Plant to safely aggregate and supply MV power to customer FCBs and the Site AUX XFMRs. All Synchronous Machines are connected to the Main Bus via a GCB, GSU, and ICB or directly to the Main Bus via a GCB. Site AUX XFMRs (and therefore Site LV E-Houses) are connected to the Main Bus via an ICB. Finally, the Customer is connected to the Main Bus via a set of FCBs.

Power System Analysis Reports and Electrical Protection Relay Configuration files shall be generated by VoltaGrid to ensure that personnel and all MV equipment are adequately protected against electrical faults and Arc Flashes. Breakers/fuses shall be coordinated such that the risk of a system-wide blackout due to a single electrical fault is minimized. The MV Voltage Distribution System customer (Master Control System) control interface is exclusively provided by SEL Electrical Protection Relays (SEL 751, SEL 700G, SEL 787, SEL 735).

A Block Diagram of a typical MV Distribution System is shown in **Figure 5** below.

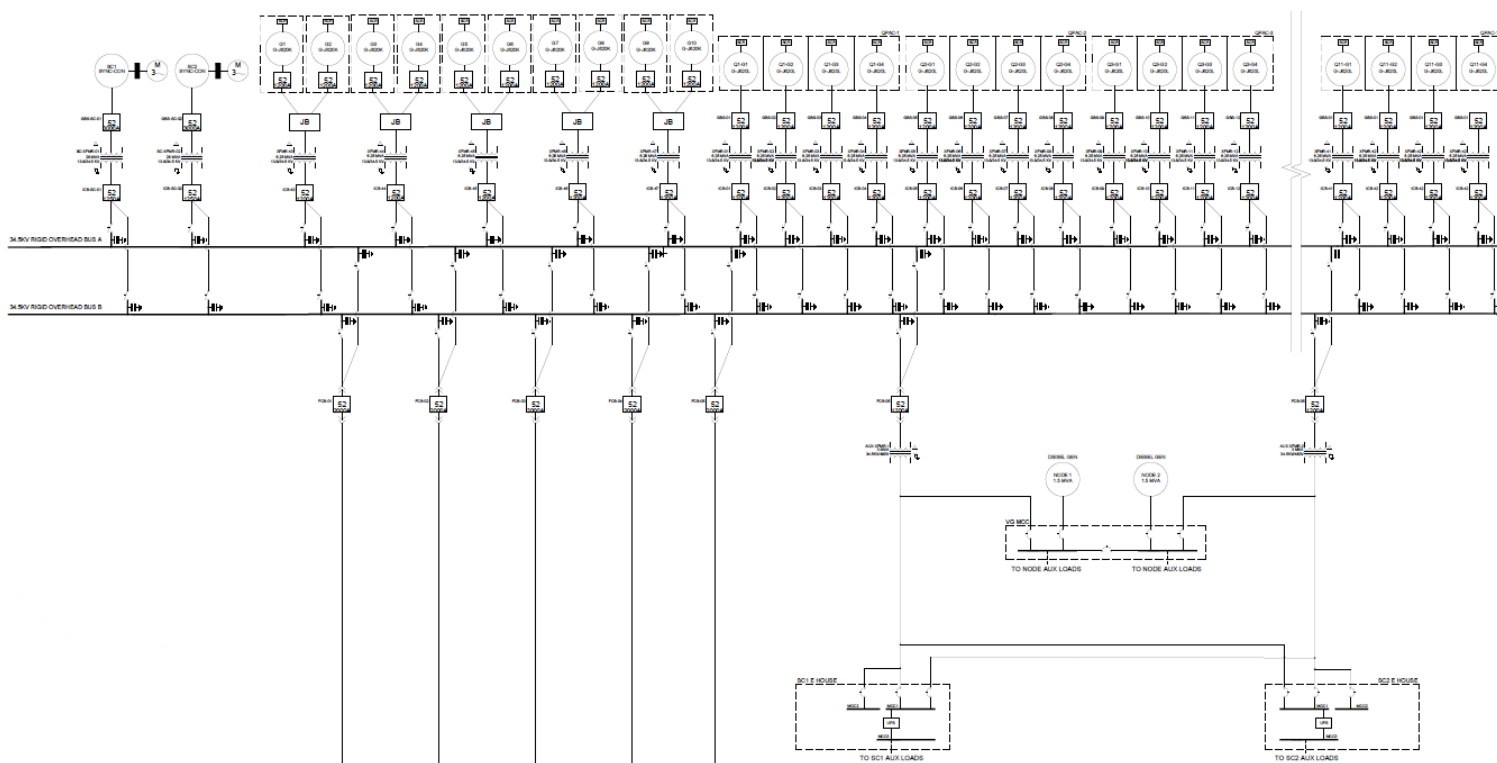


Figure 5: Typical Electrical Block Diagram

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9.1. TELEMETRIC DEVICES

The MV Distribution System shall include the following Telemetric devices:

- Breaker Open/Close Status Contacts
- Breaker Truck-Operated Contacts
- PTs
- CTs
- Alarm Contacts
- Battery Voltage Transducers

The number, type, and signals provided by these devices shall be defined by VoltaGrid, and all electrical distribution telemetric device signals shall be connected to the associated electrical protection relay (and not to any other IED including the Master Controller). All data points shall be exposed to the OT Network via SEL Relay Modbus Servers.

In addition to this telemetric data, all SEL electrical protection relays shall expose all relay word bits, alarms, and event files to the OT Network via their Modbus Servers.

9.2. ELECTROMECHANICAL CONTROL DEVICES

The Only Electromechanical Control Devices in the MV Distribution System are Trip and Close Coils. These coils are automatically actuated by SEL protection relays according to specific relay protection settings, or remotely by the Master Controller or Control Room HMI as needed to satisfy automatic control sequences or Operator control requests.

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9.3. MANUAL BACKUP CONTROL DEVICES

Remote Operation Switches (Chicken Switches) shall be used to manually open/close MV Electrical Distribution System Breakers. Remote Operation Switches shall be located at the local protection relay panel, or on/in the associated enclosure. Remote Operation Switches should be installed in a location that allows the operator to manually control SWGR without opening MV enclosures.

Remote Operation Switches shall control SWGR via one of the following methods in order of preference:

- Dry contacts connected to digital inputs at the local Electrical Protection Relay. The Electrical Protection Relay will include these inputs in its TRIP/CLOSE logic equations.
 - This method allows the electrical protection relay to monitor/record manual operations, and block-close when unsafe conditions are detected.
- Dry contacts connected directly to Trip/Close coils/electromechanical control devices.
 - This method bypasses the electrical protection relays, preventing them from monitoring manual operations, and relies on the operator to determine whether a manual close operation is safe.

Rackable SWGR shall be operated by hand (via the Racking bolt and a Racking tool). No electronic control signals are required, but Truck-Operated-Contacts should be connected to a digital Input at the electrical protection relay – allowing the Racked In/Out status to be remotely monitored (at the Control Room HMI).

Each breaker shall include manual open and close pushbuttons. These pushbuttons shall provide a dry contact connected directly to the trip/close coils. No IED signals/connections are required.

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9.4. INTELLIGENT ELECTRONIC DEVICES

MV Distribution System IEDs are exclusively SEL electrical protection relays which monitor and control MV breakers. These relays will integrate to the Master Control System (via SEL 3350 FEPs) and VoltaGrid Cluster over the GOOSE, MMS, and Modbus Ethernet communication protocols. The following sections detail the specific relays and use cases in a typical VoltaGrid Data Center Power Plant.

9.4.1. SEL 700G MV Synchronous Machine Protection Relays

Every MV Genset and DER shall be connected to a GCB protected by an SEL 700G.

The SEL 700G relay(s) shall receive the following digital and analog data from telemetric and Manual Control devices:

- **(If Required)** Generator Inlet Gas Pressure Transmitter
 - Signal type: 4-20 mA
- **(If Required)** Generator Inlet Gas Temperature Transmitter
 - Signal type: 4-20 mA
- Line-Side Phase Voltages from GCB PTs
 - Signal Type – 0-120V
- Load-Side Phase Voltages from GCB PTs
 - Signal Type – 0-120V
- Line-Side Generator Current from Genset Neutral CTs
 - Signal Type – 0-5A current
- Load-Side Generator Current from GCB CTs
 - Signal Type – 0-5A current
- Generator Neutral Voltage from GCB Neutral Grounding Resistor (NGR)
 - Signal Type – 0-240V
- Generator Neutral Current from GCB NGR CT
 - Signal Type – 0-5A current
- Breaker Closed Status Contact (52A)
 - Signal Type – NO contact
- SWGR Remote Operation Switch Trip Output
 - Signal Type – NO contact
 - **Note:** generator protection relays shall not accept remote close switch signals, as this risks unsynchronized closure – closure should only be initiated by the Genset Controller and Electrical Protection Relay after successful synchronization or dead bus arbitration sequence.

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The SEL 700G relay(s) shall send the following digital and analog control signals to electromechanical control devices and other IEDs:

- Close Breaker signal to GCB Close Coil
 - Signal Type: NO contact
 - **Note:** this NO contact should be in series with a NO contact from the Genset Controller such that the breaker will only close when the relay and Genset Controller agree that the Genset is synchronized to the Main Bus
- Open Breaker signal to GCB Trip Coil
 - Signal Type: NO contact
- Trip signal to Genset Controller
 - Signal Type: NO contact
- Overload/Short Circuit signal to Genset Controller
 - Signal Type: NO contact
- Earth Fault signal to Genset Controller
 - Signal Type: NO contact
- **(If Required)** Speed Bias (Genset/DER Real Power Load Sharing Signal) to Genset Controller
 - Signal Type: 4-20mA
- **(If Required)** Voltage Bias (Genset/DER Reactive Power Load Sharing Signal) to Genset Controller
 - Signal Type: 4-20mA

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The SEL 700G relay(s) shall be capable of sending/receiving the following control soft I/O from the Master Controller, FEPs, and/or other Electrical Protection Relays over the OT Network via an Open Source Ethernet Protocol:

- Receive: Breaker Open Request from the Master Controller/FEP
 - Protocol: GOOSE
- Receive: ICB Breaker Closed Status from the ICB SEL 787 GSU Protection Relay
 - Protocol: GOOSE
 - **Note:** Required to block-close on the GCB when the ICB is open for unsynchronized closure protection
- Receive: Differential Protection element Trip from the ICB SEL 787 GSU Protection Relay
 - Protocol: GOOSE
- Send: GCB Breaker Failure signal to the ICB SEL 787 GSU Protection Relay
 - Protocol: GOOSE
- Receive (**If Required**): Speed Bias (Genset/DER Real Power Load Sharing Signal) from the Master Controller/FEP
 - Protocol: GOOSE
- Receive (**If Required**): Voltage Bias (Genset/DER Reactive Power Load Sharing Signal) from the Master Controller/FEP
 - Protocol: GOOSE

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 700G, as needed.

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9.4.2. SEL 751 MV Generator Auxiliary Transformer Protection Relays

Every MV Genset AUX XFMR shall be connected to the Main Bus side of the GCB through an additional MV Breaker protected by an SEL 751.

The SEL 751 relay(s) shall receive the following digital and analog data from telemetric and Manual Control devices:

- AUX XFMR current from AUX XFMR CTs
 - Signal Type – 0-5A current
- AUX XFMR Secondary Voltage from secondary terminations
 - Signal Type – 0-277V (or 0-120V if PTs are available)
- Breaker Closed Status Contact (52A)
 - Signal Type – NO contact
- SWGR Remote Operation Switch Trip/Close Outputs
 - Signal Type – NO contact

The SEL 751 relay(s) shall send the following digital and analog control signals to electromechanical control devices and other IEDs:

- Close Breaker signal to GCB Close Coil
 - Signal Type: NO contact
- Open Breaker signal to GCB Trip Coil
 - Signal Type: NO contact

The SEL 751 relay(s) shall be capable of sending/receiving the following control soft I/O from the Master Controller, FEPs, and/or other Electrical Protection Relays over the OT Network via an Open Source Ethernet Protocol:

- Receive: Breaker Open Request from the Master Controller/FEP
 - Protocol: GOOSE
- Receive: Breaker Close Request from the Master Controller/FEP
 - Protocol: GOOSE
- Send: MV Breaker Failure signal to the GCB SEL 700G Genset and ICB SEL 787 GSU Protection Relays
 - Protocol: GOOSE

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 751, as needed.

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9.4.3. SEL 787 Generator Step Up Transformer Protection Relays

All GSUs shall be connected to the Main Bus via an ICB. The ICB shall be monitored and controlled by an SEL 787 XFMR protection relay.

The SEL 787 relay shall receive the following digital and analog data from telemetric and Manual Control devices:

- Line side GSU phase currents from the GGU primary side terminal box
 - Signal Type – 0-5A current
- Load-Side GGU phase currents from the GGU secondary side terminal box
 - Signal Type – 0-5A current
- GGU Neutral Current from the GGU secondary side terminal box
 - Signal Type – 0-5A current
- Breaker Closed Status Contact (52A)
 - Signal Type – NO contact
- (If Necessary) GGU Alarm contacts
 - Signal Type:
 - NO or NC contact as needed
- SWGR Remote Operation Switch Trip/Close Outputs
 - Signal Type – NO contacts

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The SEL 787 relay shall send the following digital and analog control signals to electromechanical control devices and other IEDs:

- Close Breaker signal to ICB Close Coil
 - Signal Type – NO contact
- Open Breaker signal to ICB Trip Coil
 - Signal Type – NO contact

The SEL 787 relay(s) shall be capable of sending/receiving the following control soft I/O from the Master Controller, FEPs, and/or other Electrical Protection Relays over the OT Network via an Open Source Ethernet Protocol:

- Receive: Breaker Open Request from the Master Controller/FEP
 - Protocol: GOOSE
- Receive: Breaker Close Request from the Master Controller/FEP
 - Protocol: GOOSE
- Send: ICB Breaker Closed Status to the GCB SEL 700G relay
 - Protocol: GOOSE
 - **Note:** Required for unsynchronized closure protection
- Send: ICB Differential Protection element Trip to GCB SEL 700G relay
 - Protocol: GOOSE

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 787, as needed.

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9.4.4. SEL 787 Site Auxiliary Transformer Protection Relays

All Site AUX XFMRs shall be connected to the Main Bus via an ICB. The ICB shall be monitored and controlled by an SEL 787 XFMR protection relay.

The SEL 787 relay shall receive the following digital and analog data from telemetric and Manual Control devices:

- Line side AUX XFMR phase currents from the AUX XFMR primary side terminal box
 - Signal Type – 0-5A current
- Load-Side AUX XFMR phase currents from the AUX XFMR secondary side terminal box
 - Signal Type – 0-5A current
- AUX XFMR Neutral Current from the AUX XFMR secondary side terminal box
 - Signal Type – 0-5A current
- Breaker Closed Status Contact (52A)
 - Signal Type – NO contact
- (If Necessary) AUX XFMR Alarm contacts
 - Signal Type:
 - NO or NC contact as needed
- SWGR Remote Operation Switch Trip/Close Outputs
 - Signal Type – NO contacts

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The SEL 787 relay shall send the following digital and analog control signals to electromechanical control devices and other IEDs:

- Close Breaker signal to ICB Close Coil
 - Signal Type – NO contact
- Open Breaker signal to ICB Trip Coil
 - Signal Type – NO contact

The SEL 787 relay(s) shall be capable of sending/receiving the following control soft I/O from the Master Controller, FEPs, and/or other Electrical Protection Relays over the OT Network via an Open Source Ethernet Protocol:

- Receive: Breaker Open Request from the Master Controller/FEP
 - Protocol: GOOSE
- Receive: Breaker Close Request from the Master Controller/FEP
 - Protocol: GOOSE
- Send: ICB Differential Protection element Trip to Site LV E-House Inlet SEL 751 Relay
 - Protocol: GOOSE

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 787, as needed.

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9.4.5. SEL 751 Customer Feeder Protection Relays

The Main Bus shall be coupled to the customer electrical distribution system(s) via FCBs. The FCB shall be monitored and controlled by an SEL 751 Feeder protection relay.

The SEL 751 relay shall receive the following digital and analog data from telemetric and Manual Control devices:

- Line Side Main Bus Phase Voltages
 - Signal Type: 0-120V
- Line side FCB phase currents from the FCB phase CTs
 - Signal Type – 0-5A current
- Breaker Closed Status Contact (52A)
 - Signal Type – NO contact
- SWGR Remote Operation Switch Trip/Close Outputs
 - Signal Type – NO contacts

The SEL 751 relay shall send the following digital and analog control signals to electromechanical control devices and other IEDs:

- Close Breaker signal to GCB Close Coil
 - Signal Type – NO contact
- Open Breaker signal to GCB Trip Coil
 - Signal Type – NO contact

The SEL 751 relay(s) shall be capable of sending/receiving the following control soft I/O from the Master Controller, FEPs, and/or other Electrical Protection Relays over the OT Network via an Open Source Ethernet Protocol:

- Receive: Breaker Open Request from the Master Controller/FEP
 - Protocol: GOOSE
- Receive: Breaker Close Request from the Master Controller/FEP
 - Protocol: GOOSE

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 751, as needed.

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9.4.6. SEL 735 Customer Feeder Power Meter

An SEL 735 shall be installed in every customer FCB to enable CWM and in-depth monitoring and analysis of system harmonics and transients.

The SEL 735 relay shall receive the following digital and analog data from telemetric devices:

- Line Side Main Bus Phase Voltages
 - Signal Type: 0-120V
- Line side FCB phase currents from the FCB phase CTs
 - Signal Type – 0-5A current

Additional feedback signals will be provided to the VoltaGrid Cluster and Master Controller/FEP, by the SEL 735, as needed.

9.5. FAILURE MODES

A single component failure in the Medium Voltage Distribution System should not result in a system-wide blackout unless it is a short circuit in the Main Bus or on the Main Bus side of an ICB or FCB.

Short Circuits in Medium Voltage distribution for Sync Cons may also result in system-wide blackouts depending on the Customer load profile.

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10. LOW VOLTAGE DISTRIBUTION SYSTEM

The LV Distribution System allows a typical VoltaGrid Data Center Power Plant to safely aggregate and supply LV power to all BOP loads, the Ops Building, the WIC, Site Lighting, the maintenance yard, MV Genset Blackstart Connections, Sync Con E-Houses, etc. The primary LV bus is the main 480V bus in the Site LV E-House, although other large 480V busses may be found in the system (such as Sync Con E-House MCCs 1, 2, and 3). the main 480V bus in the Site LV E-House is normally energized by the Site AUX XFMRs but, in the event of a system-wide blackout, it may also be energized by the LV Backup Genset.

Note: MV Genset AUX systems/LV power is separate from the Site LV E-House distribution system. Each MV Genset Package or its associated GCB contains a dedicated 13.8kV/480V Genset AUX XFMR which is connected on the Main Bus side of the GCB. This allows a small group of MV gensets to energize the AUX systems in all MV gensets for which the associated ICB is closed.

Power System Analysis Reports, Electrical Protection Relay Configuration files, and LV Breaker Settings shall be generated by VoltaGrid to ensure that personnel and all LV equipment are adequately protected against electrical faults and Arc Flashes. Breakers/fuses shall be coordinated such that the risk of a system-wide blackout due to a single electrical fault is minimized. The LV Voltage Distribution System customer (Master Control System) control interface is exclusively provided by SEL Electrical Protection Relays (SEL 751, SEL 700G, SEL 787, SEL 735) and SEL RTACs (SEL 2240).

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10.1. TELEMETRIC DEVICES

The LV Distribution System shall include the following Telemetric devices:

- Breaker Open/Close Status Contacts
- Breaker Truck-Operated Contacts
- PTs
- CTs
- Alarm Contacts
- Battery Voltage Transducers

The number, type, and signals provided by these devices shall be defined by VoltaGrid, and all electrical distribution telemetric device signals shall be connected to the associated breaker controls, controller, or electrical protection relay (and not to any other IED including the Master Controller). All data points shall be exposed to the OT Network via Modbus Servers.

In addition to this telemetric data, all SEL electrical protection relays shall expose all relay word bits, alarms, and event files to the OT Network via their Modbus Servers.

10.2. ELECTROMECHANICAL CONTROL DEVICES

The Only Electromechanical Control Devices in the LV Distribution System are Trip/Close actuators/coils and ATS. These coils/switches are automatically actuated by LV Distribution System controllers and SEL protection relays according to specific configurations/relay protection settings, or remotely by the Master Controller or Control Room HMI as needed to satisfy automatic control sequences or Operator control requests.

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10.3. MANUAL BACKUP CONTROL DEVICES

Remote Operation Switches (Chicken Switches) shall be used to manually open/close LV Electrical Distribution System Breakers which are rated greater than or equal to 400A. Remote Operation Switches shall be located at the local protection relay panel, or on/in the associated enclosure. Remote Operation Switches should be installed in a location that allows the operator to manually control breakers without opening enclosures or standing directly in front of 480V SWBDs.

Remote Operation Switches shall control breakers via one of the following methods in order of preference:

- Dry contacts connected to digital inputs at the local Electrical Protection Relay or RTAC. The Electrical Protection Relay/RTAC will include these inputs in its TRIP/CLOSE logic equations.
 - This method allows the electrical protection relay/RTAC to monitor/record manual operations, and block-close when unsafe conditions are detected.
- Dry contacts connected directly to Trip/Close coils/electromechanical control devices.
 - This method bypasses the electrical protection relays/RTACs, preventing them from monitoring manual operations, and relies on the operator to determine whether a manual close operation is safe.

Rackable breakers shall be operated by hand (via the Racking bolt and a Racking tool). No electronic control signals are required, but Truck-Operated-Contacts should be connected to a digital Input at the electrical protection relay/RTAC – allowing the Racked In/Out status to be remotely monitored (at the Control Room HMI).

Each breaker shall include manual open and close pushbuttons, handles, or switches. These pushbuttons/handles/switches shall provide a dry contact connected directly to the trip/close actuator. No IED signals/connections are required.

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10.4. INTELLIGENT ELECTRONIC DEVICES

LV Distribution System IEDs include SEL electrical protection relays, SEL RTACs, Breaker Controllers, and ATS Controllers which monitor and control LV breakers. These devices will integrate to the Master Control System (via SEL 3350 FEPs) and VoltaGrid Cluster over the GOOSE, MMS, and Modbus Ethernet communication protocols. The following sections detail the specific relays and use cases in a typical VoltaGrid Data Center Power Plant.

Master Control System Interfaces to devices other than SEL Electrical Protection Relays shall be provided by an SEL 2240 RTAC in the Site LV E-House. The soft I/O interface to this RTAC will be provided by the Site LV E-House OEM (NSA).

10.4.1. SEL 2240 LV E-House Controller

The SEL 2240 LV-House Controller will provide the hard-wired and/or Modbus interfaces required to control all LV breakers which are not associated with an SEL Electrical Protection Relay. It will also provide a soft I/O interface to the Site LV E-House FEPs to enable the Master Controller (and therefore the Control Room HMI) to remotely open/close and monitor these breakers.

10.4.2. SEL 751 LV E-House Feeder Breaker Protection Relays

All Site LV E-House and Sync Con E-House LV feeder breakers which are rated greater than or equal to 800A will be protected by an SEL 751.

10.4.3. SEL 700G LV E-House Tie Breaker Protection Relays

The Site LV E-House 480V busses are connected by a tie breaker. This tie breaker is protected by an SEL 700G.

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10.4.4. LV Breaker Controllers

LV Breaker controllers which require integration to the Master Control System will be integrated to the SEL 2240 LV E-House Controller.

10.4.5. Control Voltage UPS and Battery Chargers

Various UPS and Battery Chargers are included in typical VoltaGrid Data Center Power Plant designs. All UPS and Battery Charger output voltages should be provided to the Operator via the CCMS Screen in the Control Room HMI. These values may be integrated into the VoltaGrid Message Bus via a Modbus Server hosted by the UPS/Battery Charger, or by connection of its output terminals to a spare analog input on an associated SEL Electrical Protection Relay.

10.5. FAILURE MODES

A single LV Distribution System component failure should not result in a system-wide blackout. Certain Sync Con LV E-House component failures may result in a system-wide blackout depending on the customer load profile.

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11. BACKUP LOW VOLTAGE GENSETS

The LV Backup Gensets in a typical VoltaGrid Data Center Power Plant are a source of emergency LV power intended to ensure that critical AUX loads can be immediately reenergized after a system-wide blackout.

Critical AUX loads include:

- Sync Con E-House MCC 3
- Operations Building ATS
- WIC ATS
- Site Lighting
- Fuel Gas Distribution System
- Bulk Urea Distribution System

The typical LV Backup Genset used in VoltaGrid Data Center Power Plant designs is the Caterpillar C32B Genset which includes an EMCP 4.4 Local Control Panel. In addition to local, manual control interfaces, the EMCP 4.4 hosts a Modbus Server which can be used for remote monitoring and control.

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11.1. TELEMETRIC DEVICES

LV Backup Genset Packages shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Pressure switches/sensors
- Temperature switches/sensors
- Vibration sensors
- Phase CTs
- Phase PTs
- Speed sensor(s)
- Level switches/sensors
- Status Contacts (I.e. running/stopped)

The number and type of these devices and the signals provided by them shall be defined by the Genset OEM, and all Genset telemetric device signals shall be connected to the Genset Controllers or Genset sub-system Controllers.

11.2. ELECTROMECHANICAL CONTROL DEVICES

In addition to Starters, Governors, and Exciters – LV Backup Genset Packages shall include a plurality of other electromechanical control devices required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Contactors
- VFDs
- Control Valves

The number and type of these devices and their signal requirements shall be defined by the Genset OEM, and all Genset control signals shall be provided by the Genset Controllers or Genset sub-system Controllers.

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11.3. MANUAL BACKUP CONTROL DEVICES

The LV Backup Genset's Local Control Panel shall provide means to manually start and stop the Genset. This requirement is satisfied by the Caterpillar C32B Diesel Genset's EMCP 4.4 Control Panel.

11.4. INTELLIGENT ELECTRONIC DEVICES

In addition to Genset, Engine, and Excitation Controllers – LV Backup Genset Packages shall include a plurality of other IEDs required for the safe, effective, and efficient operation and supervision of the Genset package including but not limited to the following:

- Contactors
- VFDs
- Control Valves
- Electronically Controlled Clutches

The number, type, and signal requirements of these devices shall be defined by the LV Backup Genset OEM, and all Genset package control signals shall be connected to the Genset Controller or Genset sub-system controller. All control signal data points shall be exposed to the OT Network via an Open Source Ethernet Protocol.

The LV Backup Genset Controller shall provide/expose a hard or soft I/O customer control interface allowing the Master Controller (and therefore the Control Room HMI) to remotely start and stop the LV Backup Genset. The Interface specification will be provided by the LV Backup Genset OEM.

The typical remote control interface for the CAT EMCP 4.4 is a Modbus Server, and Caterpillar will provide the Modbus Register Map.

11.5. FAILURE MODES

An LV Backup Genset failure shall not result in a system-wide blackout. The LV Backup Genset should allow the plant to safely and efficiently reenergize after a system-wide blackout.

12. FUEL GAS DISTRIBUTION SYSTEM

The Fuel Gas Distribution system distributes fuel to all Genset Packages via a set of redundant central manifolds. A typical Fuel Gas Distribution System PFD is shown below in **Figure 6**.

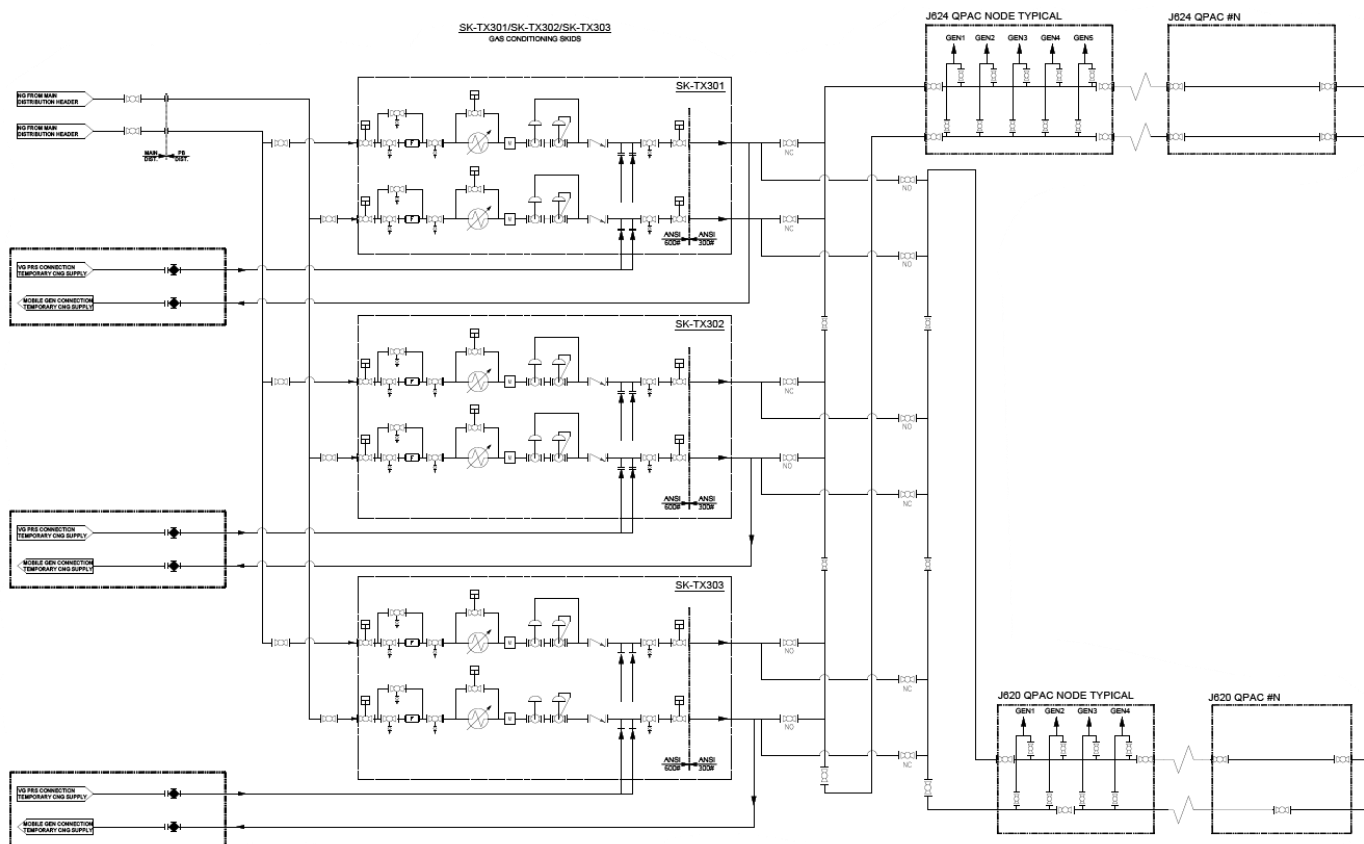


Figure 6: Typical Fuel Gas Distribution System PFD

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12.1. TELEMETRIC DEVICES

The Fuel Gas Distribution System shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the plant including but not limited to the following:

- Pressure sensors
- Temperature sensors
- Flow Meter(s)
- Valve Position contacts
- LEL switches/sensors
- Fire Alarm switches/sensors

The number and type of these devices and the signals provided by them shall be defined by the Fuel Gas Distribution System OEM, and most Fuel Gas Distribution System telemetric device signals shall be connected to the Fuel Gas Distribution System Controllers.

Some Fuel Gas Distribution System telemetric device signals may be connected to a centralized BOP Remote I/O Module (SEL 2240 or equivalent). The centralized BOP Remote I/O Module shall expose these signals to the OT Network via a Modbus Server. The associated Modbus Register Map shall be developed by VoltaGrid.

Fuel pressure/temperature sensors at the individual genset level shall be connected to analog inputs at the Gensets SEL 700G Generator Protection Relay. These signal values shall be exposed to the OT Network via the SEL 700G relay's Modbus Server. The SEL 700G Modbus Register Map is provided in its Instruction Manual.

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12.2. ELECTROMECHANICAL CONTROL DEVICES

The Fuel Gas Distribution System shall include a plurality of electromechanical control devices required for the safe, effective, and efficient operation of the system including but not limited to the following:

- Contactors
- Control Valves

The number and type of these devices and their signal requirements shall be defined by the Fuel Gas Distribution System OEM, and all Fuel Gas Distribution System control signals shall be provided by the Fuel Gas Distribution System Controllers.

12.3. MANUAL BACKUP CONTROL DEVICES

Hand-Operated valves and local control panels/HMIs may or may not be included in the Plant Control System as needed for start-up and emergency operation procedures related to the Fuel Gas Distribution System.

The Plant Control System shall include strategically positioned Emergency Shut Down Buttons located throughout the Power Plant which are included in a series, fail-safe, circuit connected to the Fuel Gas Distribution System's ESDV.

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12.4. INTELLIGENT ELECTRONIC DEVICES

The Fuel Gas Distribution System Controllers should work independently from the Master Controller to automatically monitor and control all Gas Distribution primary and AUX systems. These systems include but are not limited to the following:

- Heating Systems
- Safety Systems
- Monitoring Systems
- Pressure Reduction Systems
- Power Distribution Systems

The Fuel Gas Distribution System Controllers may outsource any/all metering and control signals to sub-system controllers or remote I/O modules, however most Fuel Gas Distribution System telemetric and control signal values should be aggregated in a central Modbus Server in the main Fuel Gas Distribution System Controllers.

The number and type of inputs/outputs to/from telemetric devices, electromechanical control devices, and the Fuel Gas Distribution System Controllers shall be defined by the Fuel Gas Distribution System OEM. All Fuel Gas Distribution System telemetry/control signals collected by the Fuel Gas Distribution System Controller shall be exposed to the OT Network via a Modbus Server. The Fuel Gas Distribution System OEM shall provide VoltaGrid with the associated Modbus Register Map.

Note: As described in [Telemetric Devices](#), some Fuel Gas Distribution System telemetry will be exposed to the OT Network via the BOP Remote I/O Module (SEL 2240 or equivalent) and Generator Protection Relays (SEL 700G or equivalent) Modbus Servers.

12.5. FAILURE MODES

A single Fuel Gas Distribution System component failure shall not result in a system-wide blackout. Redundant Algas skids and/or distribution manifolds are included in the system design to ensure continuous availability of the Fuel Gas Distribution System.

13. BULK UREA DISTRIBUTION SYSTEM

The Bulk Urea Distribution System distributes Urea to the day tanks in all Genset SCR Packages via a set of redundant pumps and central manifolds. A typical Bulk Urea Distribution System PFD is shown below in **Figure 7**.

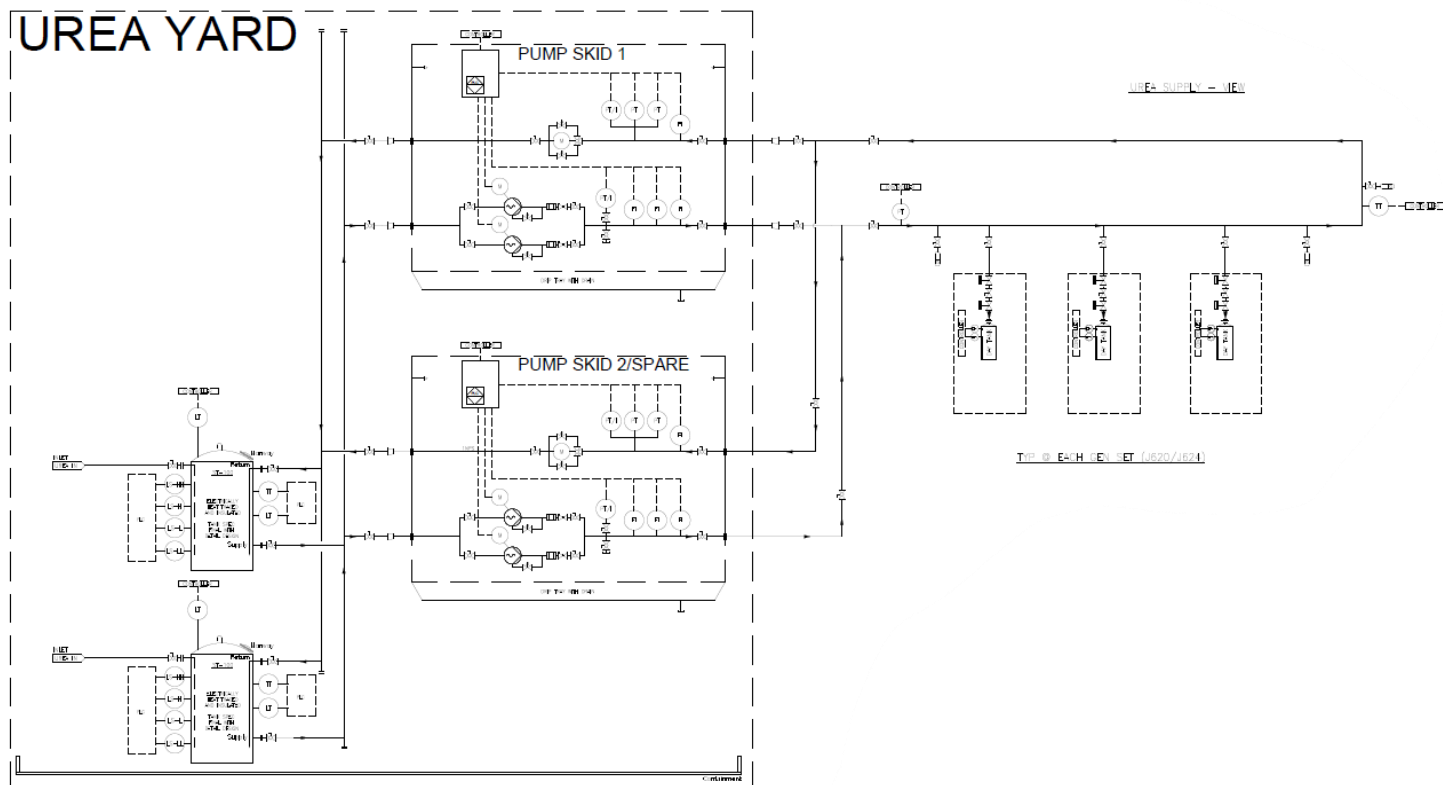


Figure 7: Bulk Urea Distribution System PFD

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13.1. TELEMETRIC DEVICES

The Bulk Urea Distribution System shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the plant including but not limited to the following:

- Pressure sensors
- Temperature sensors
- Flow Meter(s)
- Level switches/sensors
- Valve Position contacts
- Fire Alarm switches/sensors

The number and type of these devices and the signals provided by them shall be defined by the Bulk Urea Distribution System OEM, and most Bulk Urea Distribution System telemetric device signals shall be connected to the Bulk Urea Distribution System Controllers.

Some Bulk Urea Distribution System telemetric device signals may be connected to a centralized BOP Remote I/O Module (SEL 2240 or equivalent). The centralized BOP Remote I/O Module shall expose these signals to the OT Network via a Modbus Server. The associated Modbus Register Map shall be developed by VoltaGrid.

13.2. ELECTROMECHANICAL CONTROL DEVICES

The Bulk Urea Distribution System shall include a plurality of electromechanical control devices required for the safe, effective, and efficient operation of the system including but not limited to the following:

- Contactors
- Control Valves

The number and type of these devices and their signal requirements shall be defined by the Bulk Urea Distribution System OEM, and all Bulk Urea Distribution System control signals shall be provided by the Bulk Urea Distribution System Controllers.

13.3. MANUAL BACKUP CONTROL DEVICES

Hand-Operated valves and local control panels/HMIs may or may not be included in the Plant Control System as needed for start-up and emergency operation procedures related to the Bulk Urea Distribution System.

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13.4. INTELLIGENT ELECTRONIC DEVICES

The Bulk Urea Distribution System Controller(s) should work independently from the Master Controller to automatically monitor and control all Bulk Urea Distribution System primary and AUX systems. These systems include but are not limited to the following:

- Pump Skid Systems
- Flow Control Systems
- Monitoring Systems
- Power Distribution Systems

The Bulk Urea Distribution System Controllers may outsource any/all metering and control signals to sub-system controllers or remote I/O modules, however most Bulk Urea Distribution System telemetric and control signal values should be aggregated in a central Modbus Server in the main Bulk Urea Distribution System Controllers.

The number and type of inputs/outputs to/from telemetric devices, electromechanical control devices, and the Bulk Urea Distribution System Controllers shall be defined by the Bulk Urea Distribution System OEM. All Bulk Urea Distribution System telemetry/control signals collected by the Bulk Urea Distribution System Controller shall be exposed to the OT Network via a Modbus Server. The Bulk Urea Distribution System OEM shall provide VoltaGrid with the associated Modbus Register Map.

Note: As described in [Telemetric Devices](#), some Bulk Urea Distribution System telemetry will be exposed to the OT Network via the BOP Remote I/O Module (SEL 2240 or equivalent) Modbus Server.

13.5. FAILURE MODES

A single Bulk Urea Distribution System component failure shall not result in a system-wide blackout. Redundant pump skids and distribution manifolds are included in the system design to ensure continuous availability of the Bulk Urea Distribution System.

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14. SELECTIVE CATALYTIC REDUCTION PACKAGES

SCR Packages are typically connected to the exhaust systems of all MV Gensets in VoltaGrid Data Center Power Plants. The SCRs allow the MV Gensets to satisfy emissions permit requirements by dosing the exhaust gas with Urea, from the Bulk Urea Distribution System, to reduce nitrogen oxide emissions.

14.1. TELEMETRIC DEVICES

SCR Packages shall include a plurality of telemetric devices required for the safe, effective, and efficient operation and supervision of the plant including but not limited to the following:

- Pressure sensors
- Temperature sensors
- Exhaust Gas Composition sensor(s)
- Level switches/sensors
- Valve Position contacts
- Fire Alarm switches/sensors

The number and type of these devices and the signals provided by them shall be defined by the SCR or Genset OEM, and all SCR telemetric device signals shall be connected to the SCR Controllers or Genset Controllers.

14.2. ELECTROMECHANICAL CONTROL DEVICES

The SCR Packages shall include a plurality of electromechanical control devices required for the safe, effective, and efficient operation of the system including but not limited to the following:

- Contactors
- Control Valves
- Dosing Valves

The number and type of these devices and their signal requirements shall be defined by the SCR OEM or Genset OEM, and all SCR control signals shall be provided by the SCR Controllers or Genset Controllers.

14.3. MANUAL BACKUP CONTROL DEVICES

Hand-Operated valves and local control panels/HMIs may or may not be included in the Plant Control System as needed for start-up and emergency operation procedures related to the SCR Packages.

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14.4. INTELLIGENT ELECTRONIC DEVICES

The SCR Controllers or Genset Controllers should work independently from the Master Controller to automatically monitor and control all SCR primary and AUX systems. These systems include but are not limited to the following:

- Dosing Systems
- Day Tank Management Systems
- Emissions Monitoring Systems
- Power Distribution Systems

The SCR Controller or Genset Controller may outsource any/all metering and control signals to sub-system controllers or remote I/O modules, however all SCR telemetric and control signal values should be aggregated in a central Modbus Server in the main SCR Controller or Genset Controller.

The number and type of inputs/outputs to/from telemetric devices, electromechanical control devices, and the SCR Controllers or Genset Controllers shall be defined by the SCR and/or Genset OEM. All SCR telemetry/control signals collected by the SCR Controller or Genset Controller shall be exposed to the OT Network via a Modbus Server. The SCR or Genset OEM shall provide VoltaGrid with the associated Modbus Register Map.

14.5. FAILURE MODES

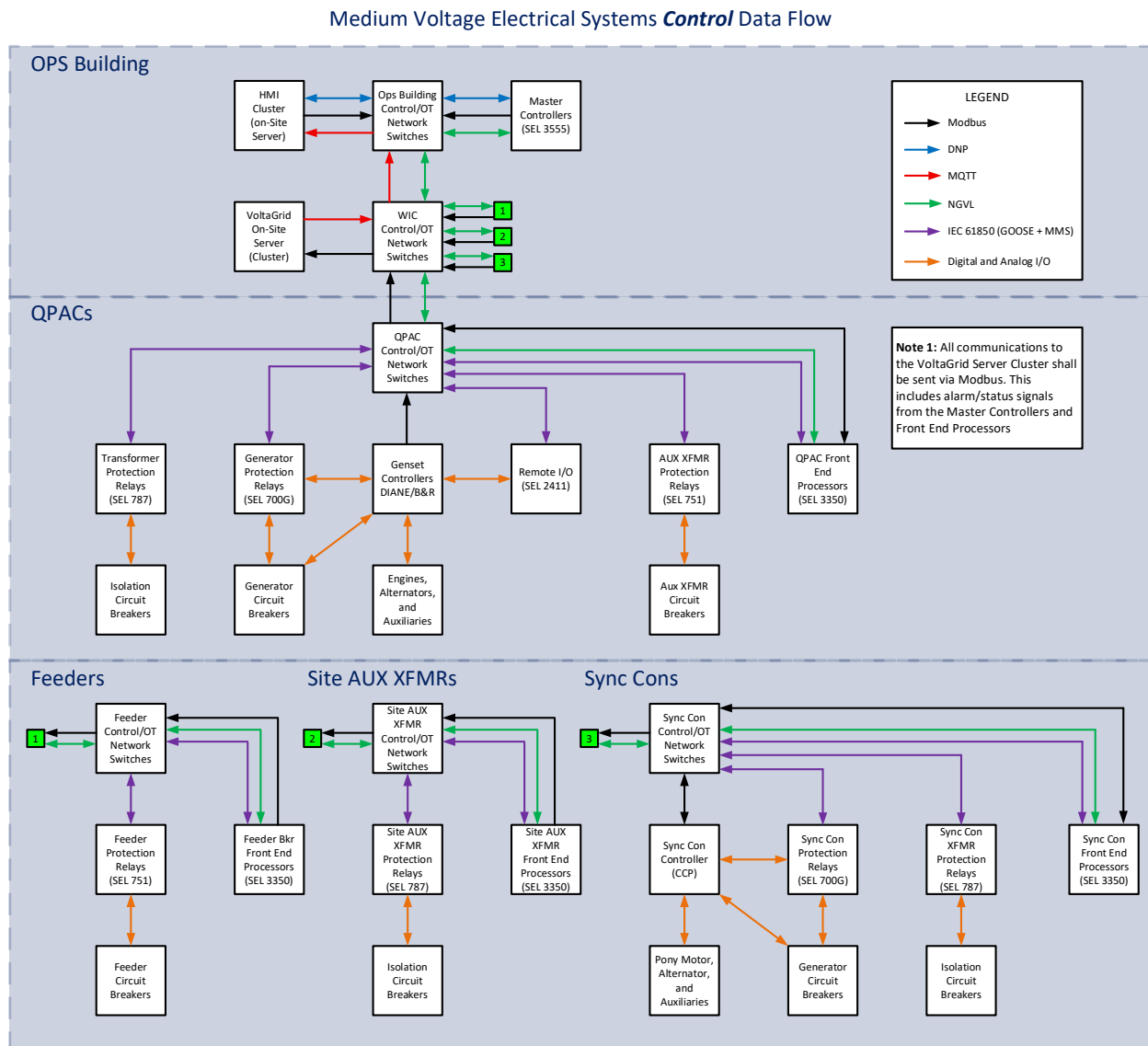
A single SCR Package failure shall not result in a system-wide blackout. Individual SCR failures may trigger individual engine shutdowns if required.

Note: The Master Controller's IRM Management control sequence should maintain a minimum of N+2 gensets synchronized to the Main Bus – preventing an individual engine shut down from causing a system-wide blackout.

15. MASTER CONTROLLER

The Plant Control System shall include redundant Master Controllers designed to automatically monitor and control the Power Plant. The Master Controllers should also execute Operator control requests from the Control Room HMI.

In general, the Master Controllers can and should integrate with individual asset Hard and Soft I/O based control interfaces over Open Source Ethernet Protocols, FEPs, and Remote I/O Modules. The typical data flows between the Master Controllers (SEL 3555), FEPs (SEL 3350), Remote I/O Modules (SEL 2411, SEL 2240, SEL 700G, SEL 751, SEL 787, etc.), and MV/LV equipment IEDs are shown in **Figure 8 and Figure 9**.



Low Voltage Electrical Systems *Control* Data Flow

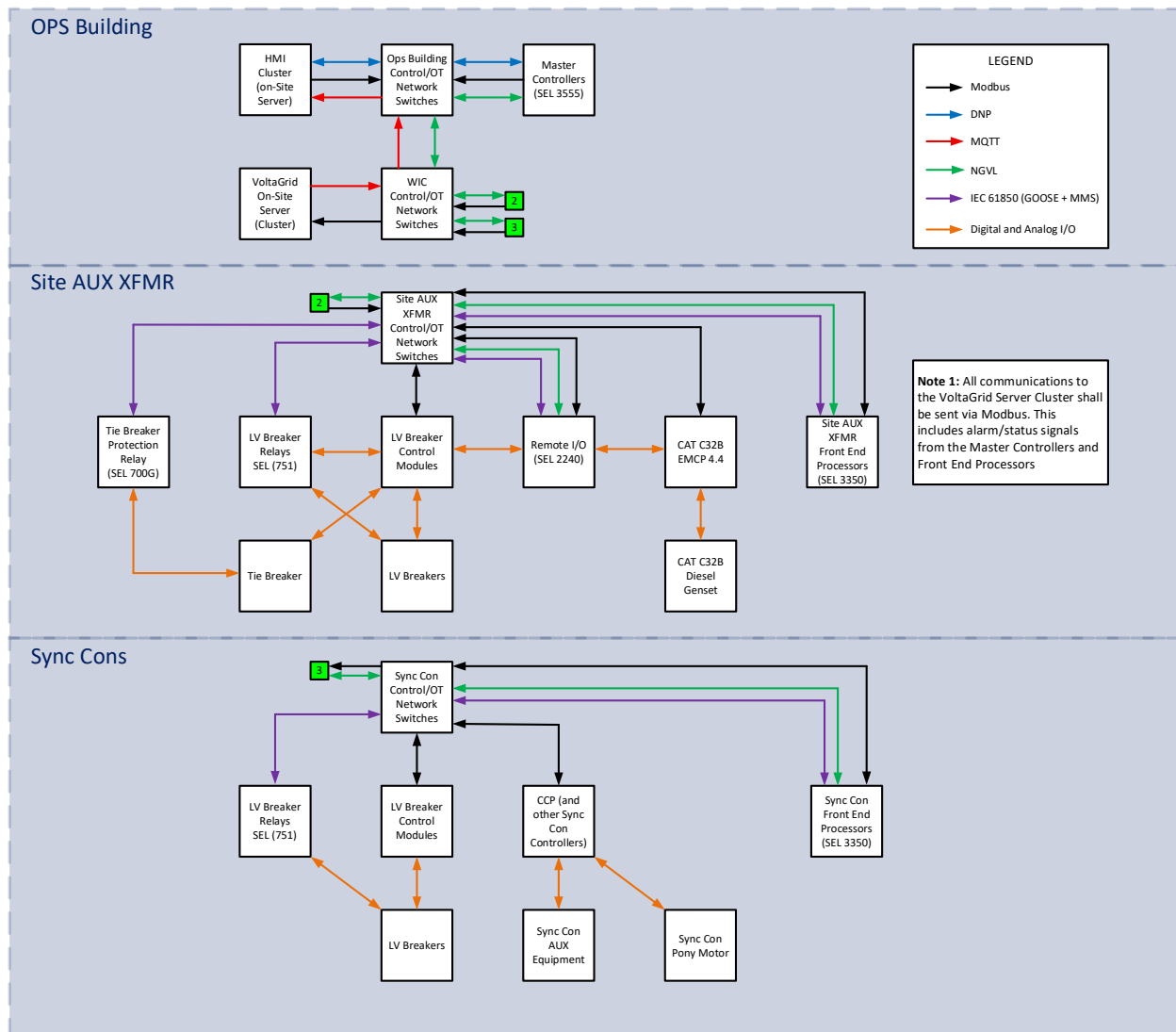


Figure 9: Low Voltage Electrical Systems Control Signal Flow Diagram

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15.1. MEDIUM VOLTAGE GENSET CONTROLLER INTERFACE

The Master Controller (SEL 3555) can and should access the MV Genset customer control interfaces and required feedback signals via Ethernet-based messages (NGVL) messages to/from an FEP (SEL 3350). The FEP will then send/receive NGVL, GOOSE, MMS, and/or Modbus messages to/from the Genset Controller, Remote I/O Module (SEL 2411), and/or Generator Protection Relay (SEL 700G) as needed to execute the Master Controller commands and provide feedback signals to the Master Controller.

MV Genset customer control interface shall be defined and provided by the MV Genset OEM. All hardwired interfaces between the MV Genset Controller and the Master Control System can and should be satisfied by the Remote I/O Modules and Generator Protection Relays.

15.2. MEDIUM VOLTAGE DER CONTROLLER INTERFACE

The Master Controller (SEL 3555) can and should access the DER customer control interfaces and required feedback signals via Ethernet-based messages (NGVL) messages to/from an FEP (SEL 3350). The FEP will then send/receive NGVL, GOOSE, MMS, and/or Modbus messages to/from the DER Controller, Remote I/O Module (SEL 2411), and/or Generator Protection Relay (SEL 700G) as needed to execute the Master Controller commands and provide feedback signals to the Master Controller.

DER customer control interface shall be defined and provided by the DER OEM. All hardwired interfaces between the MV Genset Controller and the Master Control System can and should be satisfied by the Remote I/O Modules and Generator Protection Relays.

This interface is likely to be satisfied by a Modbus Server in an ABB AMS 1400 CCP.

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15.3. MEDIUM VOLTAGE DISTRIBUTION INTERFACE

The Master Controller can and should access the MV Distribution System customer control interfaces and required feedback signals via Ethernet-based messages (NGVL) messages to/from an FEP. The FEP will then send/receive NGVL, GOOSE, and MMS messages to/from electrical protection relays (SEL 700G, SEL 751, and SEL 787) as needed to execute the Master Controller commands and provide feedback signals to the Master Controller.

VoltaGrid will provide the Master Controller OEM with drawings detailing the specific electrical protection relay part numbers and hard-wired I/O configurations.

The function of these interfaces is to enable the Master Controller (and therefore the Control Room HMI) to open/close MV Breakers, execute automatic control sequences, and monitor the state of the MV Distribution System.

The Master Control System should be able to open/close the following breakers:

- GCBs
 - Open Command only – Close Command is locally-automatically controlled by the Genset Controller and SEL 700G
 - Close request may be inadvertently controlled by the Master Controller via a Start Command when remote control is enabled at the Genset Controller – See [Manual Backup Control Devices](#).
- Genset MV AUX XFMR Breakers
- ICBs (for Gensets, DERs, and Site AUX XFMRs)
- FCBs

All hard-wired interfaces between the MV Breakers (and their associated telemetric and electromechanical control devices) and the Master Control System can and should be satisfied by the electrical protection relays.

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15.4. LOW VOLTAGE DISTRIBUTION SYSTEM INTERFACE

The Master Controller can and should access the LV Distribution System customer control interfaces and required feedback signals via Ethernet-based messages (NGVL) messages to/from an FEP. The FEP will then send/receive NGVL, GOOSE, and MMS messages to/from Site LV E-House Controllers (SEL 2240), and/or electrical protection relays (SEL 751/SEL 700G) as needed to execute the Master Controller commands and provide feedback signals to the Master Controller.

The Site LV E-House Controller configuration and Soft I/O customer control/feedback interface will be provided by the Site LV E-House OEM (NSA). This controller will be integrated with all electrical protection relays and breaker controls required to remotely operate and monitor the Site LV E-House.

Sync Con CCP and electrical protection relay Soft I/O customer control/feedback interfaces will be provided by the Sync Con OEM (ABB). The CCP and electrical protection relays will be integrated with all breaker controls required to remotely operate and monitor the Sync Con E-House

The function of these interfaces is to enable the Master Controller (and therefore the Control Room HMI) to open/close LV Breakers, execute automatic control sequences, and monitor the state of the LV Distribution System.

The Master Controller should be able to open/close all LV breakers in the Site-Specific SLD which are rated for greater than or equal to 200A.

All hardwired interfaces between the LV Breakers and the Master Control System can and should be satisfied by the Site LV E-House Controller and electrical protection relays.

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15.5. BACKUP LOW VOLTAGE GENSET CONTROLLER INTERFACE

The Master Controller can and should access the Backup LV Genset control interfaces and required feedback signals via Ethernet-based messages (NGVL) messages to/from an FEP. The FEP will then send/receive NGVL, GOOSE, MMS, and/or Modbus messages to/from the Genset Controller, and/or electrical protection relays (SEL 751/SEL 700G) as needed to execute the Master Controller commands and provide feedback signals to the Master Controller.

The function of these interfaces is to enable the Master Controller (and therefore the Control Room HMI) to start/stop the Backup LV genset Breakers, execute automatic control sequences, and monitor the state of the Backup LV Genset.

All hardwired interfaces between the Backup LV Genset and the Master Control System can and should be satisfied by the Site LV E-House Controller and electrical protection relays.

The Backup LV Genset Controller OEM shall provide the hard and/or soft I/O interfaces required to remotely start, synchronize, stop, and monitor the Backup LV Genset.

This interface is likely to be satisfied by a Modbus Server in a Caterpillar EMCP 4.4 Genset Control Panel.

15.6. MOBILE J620 GENSET TRAILER INTERFACE (FUTURE)

Requirements are to be determined by INNIO, SEL, and VoltaGrid.

15.7. MESSAGE BUS AND CONTROL ROOM HMI INTERFACE

The Master Controller shall provide a Soft I/O interface to the VoltaGrid Message Bus and Control Room HMI which enables the Operator Controls described in [Control Room Human Machine Interface](#) and provides all necessary Alarms and Alerts to VoltaGrid Message Bus.

It is preferred that this Soft I/O Interface is established using a Modbus Server hosted by the Master Controller. The Master Controller OEM shall provide a Modbus Register Map for this server. The Ignition project file for the Control Room HMI will contain a Modbus client which writes to the Master Controller holding registers in order to execute Operator Control Requests.

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15.8. LOCAL AND REMOTE MV GENSET OPERATING MODES

The local MV Genset HMIs and the Control Room HMI provide the Operator with the ability to select operating modes for each MV Genset. These two sets of selections are described in Section 7.3 [Manual Backup Control Devices](#) and Section 17.4 [Control Room Human Machine Interface](#) and they should not be confused with one-another although they have similar/overlapping functions.

The Master Controller should interpret local genset HMI and Control Room HMI Genset operating mode selections as listed below (in order of priority):

- Operator Panel Operation Selector = Off
 - Genset is locally Blocked
- Control Room HMI Mode Selector = Block
 - Genset is Remotely Blocked
 - Master Controller should drive the “Remote Block” digital Output on the Remote I/O Module (SEL 2411) high
- Operator Panel Operation Selector = Man
 - Genset is in local control mode and not available for remote start/stop
- Operator Panel Synch Selector = Off
 - Genset synchronization is locally Blocked
- Operator Panel Synch Selector = Man
 - Genset is in local control mode and not available for remote start/stop
- Operator Panel Request Selector = Off
 - Genset is in local control mode and not available for remote start/stop
- Operator Panel Request Selector = Off
 - Genset is in local control mode and not available for remote start/stop
- Operator Panel Operation Selector = Aut, Request Selector = Remote, and Synch Selector = Aut
 - Genset is available for remote start/stop
- Control Room HMI Mode Selector = Manual
 - Genset Start/Stop from the Control Room HMI is enabled, but automatic starting/stopping for IRM, Party Mode, and Minimum Load management is disabled
- Control Room HMI Mode Selector = Auto
 - Genset is available for automatic starting/stopping as required for IRM, Party Mode, and Minimum Load management

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15.9. SPEED/VOLTAGE CONTROL AND LOAD SHARING

During normal operations, the Master Control System (not necessarily the Master Controller) should modulate the Voltage and Speed Setpoints at individual MV Genset Controllers and DER Controllers to balance the real and reactive power loads across all generators and to utilize DERs to minimize voltage/frequency disturbances as much as possible. If required, Speed and voltage bias signals should be transmitted from the Master Control System to the individual Genset and/or DER Controllers via an Open Source Ethernet Protocol over redundant OT Networks.

The Genset/DER Controllers and Master Controller shall run all gensets/DERs in a communication-assisted Droop Load Sharing mode via speed/voltage bias signals from the Master Controller to maintain nominal frequency and voltage during steady-state operations.

Note: If Prudent Synchronous Condenser CCPs may be configured to run their Excitation Control Systems in Constant Voltage mode.

Note: The Master Controller may outsource Speed/Voltage Bias Control signal modulation/processing/updates to an FEP, remote I/O module, or generator protection relay, however it should continue to monitor the Genset's Frequency/Voltage stability and take corrective actions if erratic behavior is observed.

Note: Extreme care should be taken to ensure Under-Frequency Roll Off, Under Frequency Knee Point, and/or V/Hz slope settings are consistent/aligned across all Genset and DER controllers to prevent kVAR overloading of specific units during transient events.

15.10. CONSTANT P/Q CONTROL MODES AND SET POINTS (FUTURE)

Functional requirements to be determined by VoltaGrid.

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15.11. GENERATOR PRIORITY MANAGEMENT

The Master Controller shall provide soft I/O inputs allowing the Operator to set an independent priority level for each genset in the plant. The Operator should be able to set each genset to a different priority level or set group of gensets to the same priority level. For example, if there are 50 gensets in a plant, the Operator should be able to set them to priority levels 1 through 50 or split them into groups of 5 with 10 priority levels or groups of 10 with 5 priority levels.

These priority levels will be used to determine which genset should be started/stopped in pursuance of the IRM and Minimum Load Management control algorithms defined in the sections below.

If there are multiple gensets in the same priority level, and the Master controller has to determine which one to start/stop on another parameter, then one of the gensets with that priority level should be selected based on engine hours. The unit with the most engine hours should be selected for stops, and the unit with the least engine hours should be selected for starts.

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15.12. INCREMENTAL RESERVE MARGIN MANAGEMENT

During normal operations, the Master Controller should attempt to ensure that a minimum number of gensets are synchronized to the bus as selected by the Operator in a “Minimum Synchronized Generators” setting. This setting should be accessible from the Control Room HMI.

Additionally, the Master Controller should attempt to maintain enough generators (rounded up) synchronized to the bus to support a configurable percentage of the customer load. The percentage should be selected by the Operator in a “Incremental Reserve Margin” setting. This setting should be accessible from the Control Room HMI. i.e. if the customer load is 30 MW, the Operate has set IRM to 150%, and the plant’s MV Gensets are all 3.3 MW J620s, then 14 gensets should be synchronized to the bus as shown below:

$$\frac{30 \cdot 1.5}{3.3} = 13.63 \rightarrow 14$$

If the IRM Management and the Minimum Load Management control algorithms are in conflict, IRM Management shall take precedence over Minimum Load Management.

The Master Controller may automatically start any generators which are set to “Auto” from the Control Room HMI in order to satisfy either of the objectives defined above. Whenever the Master Controller deems it necessary to automatically start a genset, it should start two gensets. If both gensets succeed in synchronizing to the bus, the Master Controller may shut a unit down in order to satisfy the Minimum Load Management control algorithm provided that the engine shut down would not cause an IRM Management violation.

Automatic Genset starts should always be sent to Genset controllers according to Genset Priority Settings – Highest priority (1, 2, 3, etc.) should be started first.

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15.13. MINIMUM LOAD MANAGEMENT

During normal operations, the Master Controller should attempt to ensure that all gensets are at minimum load percentage as selected by the Operator in a “Minimum Genset Load” setting. This setting should be accessible from the Control Room HMI. If shutting a genset down would not violate the IRM requirements defined above, and the average genset load is below the Minimum Genset Load setting, then the Master Controller should shut down the lowest priority Genset which is set to Auto in the Control Room HMI.

If the IRM Management and the Minimum Load Management control algorithms are in conflict, IRM Management shall take precedence over Minimum Load Management.

If an INNIO J620/J624 DIANE Genset Controller asserts the “Delayed Shutdown/Stop/Trip Request” signal, the Master Controller should preemptively start gensets as required to maintain IRM requirements.

Automatic Genset stops should always be sent to Genset controllers according to Genset Priority Settings – Lowest priority (100, 99, 98, etc.) should be stopped first.

15.14. PARTY MODE

When the Operator sets the “Party Mode” toggle in the Control Room HMI to “On”, the Master Controller should start all gensets which are enabled for automatic starting as defined in [Manual Backup Control Devices](#) and [Control Room Human Machine Interface](#). At the same time, the Master Controller should disable the IRM and Minimum Load Management control sequences until Party Mode is turned off.

Note: In the future, Party Mode may be automatically actuated by the Data Center Owner/Operator/Control System via an interface to the VoltaGrid Cluster.

15.15. FAST DISCONNECT

In the event of a GCB Breaker Failure, as detected by the DIANE (GCB Opening Failure signal) or the SEL 700G (52BF Relay word bit), the Master Control System should drive the DIANE “Fast Disconnect” digital input high – immediately shutting the engine down.

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15.16. BLACKOUT RECOVERY AND BLACK START

In the event of a system-wide Blackout, the Master Controller should automatically initiate a Blackout Recovery control sequence and then, once the Blackout Recovery is complete or exited, allow the Operator to initiate an Automatic Black Start control sequence. The sequence of operations for Blackout Recovery and Black Start are shown below in **Figure 10** and **Figure 11**.

If either of these automatic sequences stop due to abnormal conditions or a controls failure, the Control Room HMI should provide a popup with a description of the issue and buttons allowing the Operator to Continue, Restart, or Exit the sequence.

Critical AUX Loads are as follows:

- Sync Con E-House MCC 3
- Operations Building ATS
- WIC ATS
- Site Lighting
- Fuel Gas Distribution System
- Bulk Urea Distribution System

These loads are required for the safe shut down of the Synchronous Condenser (if required), situational awareness, and the safe performance of the Black Start sequence.

The first MV Genset to start after a Blackout should start, and closed into the main bus, under-excited in order to soft start its GSU and the Synchronous Condenser GSU. Then Genset Controller can do this automatically whenever the Main Bus is deenergized, or the Master Controller can request this function via remote I/O (SEL 2411) and a digital input at the Genset Controller.

Once the Main Bus is reenergized, the Master Controller should disable IRM and Minimum Load Management then automatically start and synchronize a number of gensets selected by the Operator in a “Number of Secondary Black Start Generators” setting. This setting should be accessible from the Control Room HMI.

Once the Black Start Sequence is complete the Master Controller should reenale IRM and Minimum Load Management.

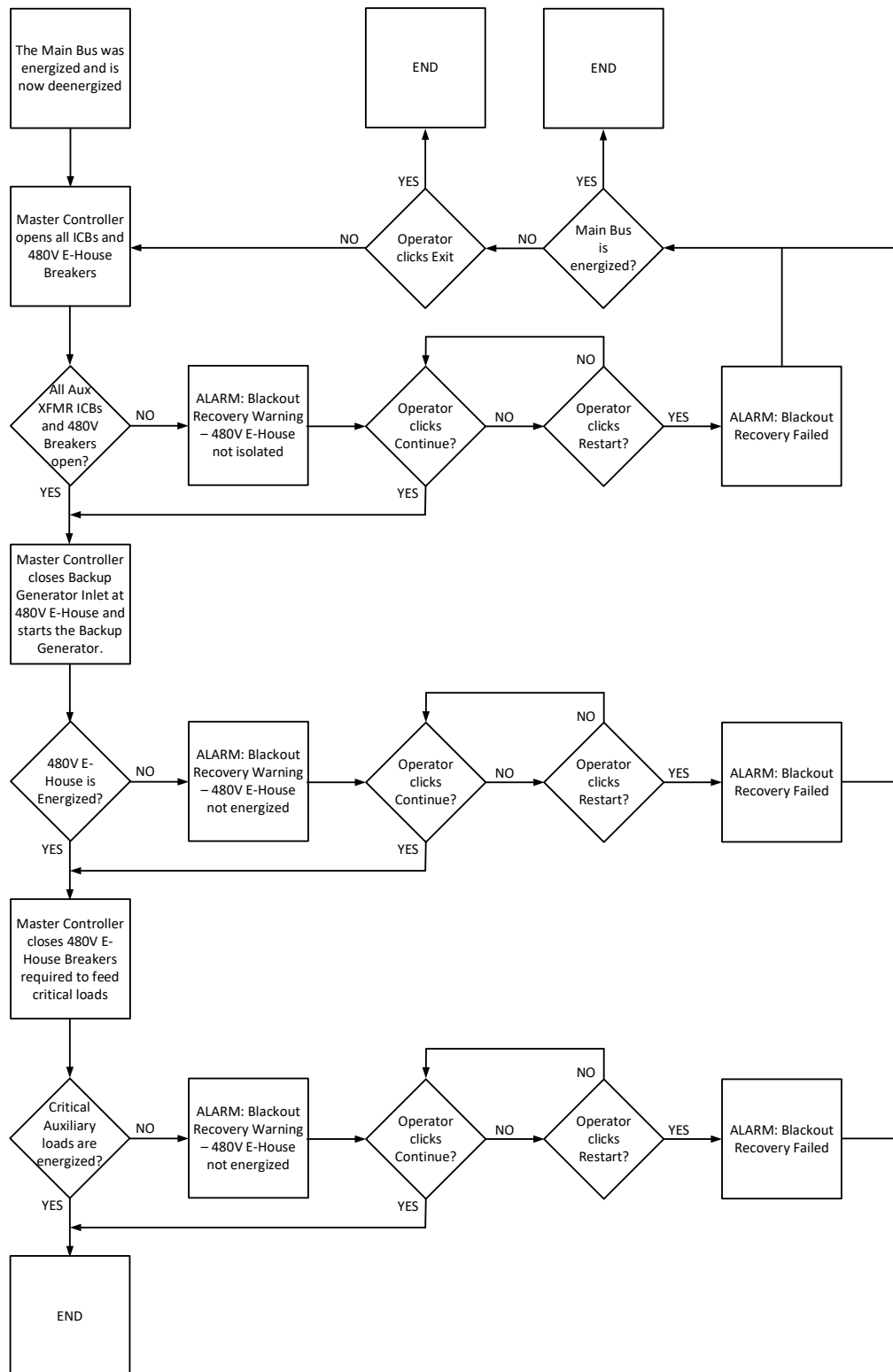


Figure 10: Blackout Recovery Sequence of Operations

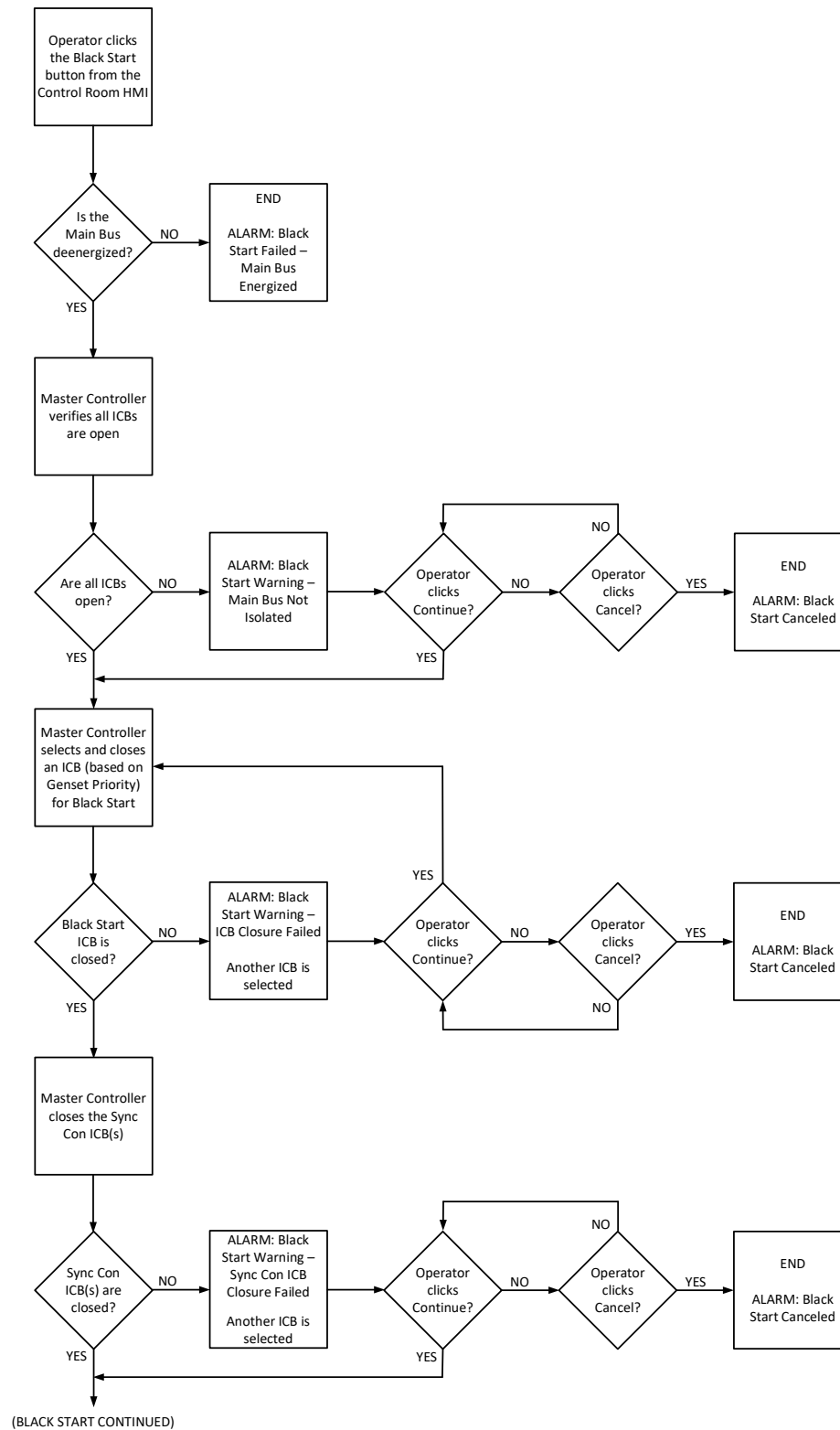


Figure 11: Black Start Sequence of Operations

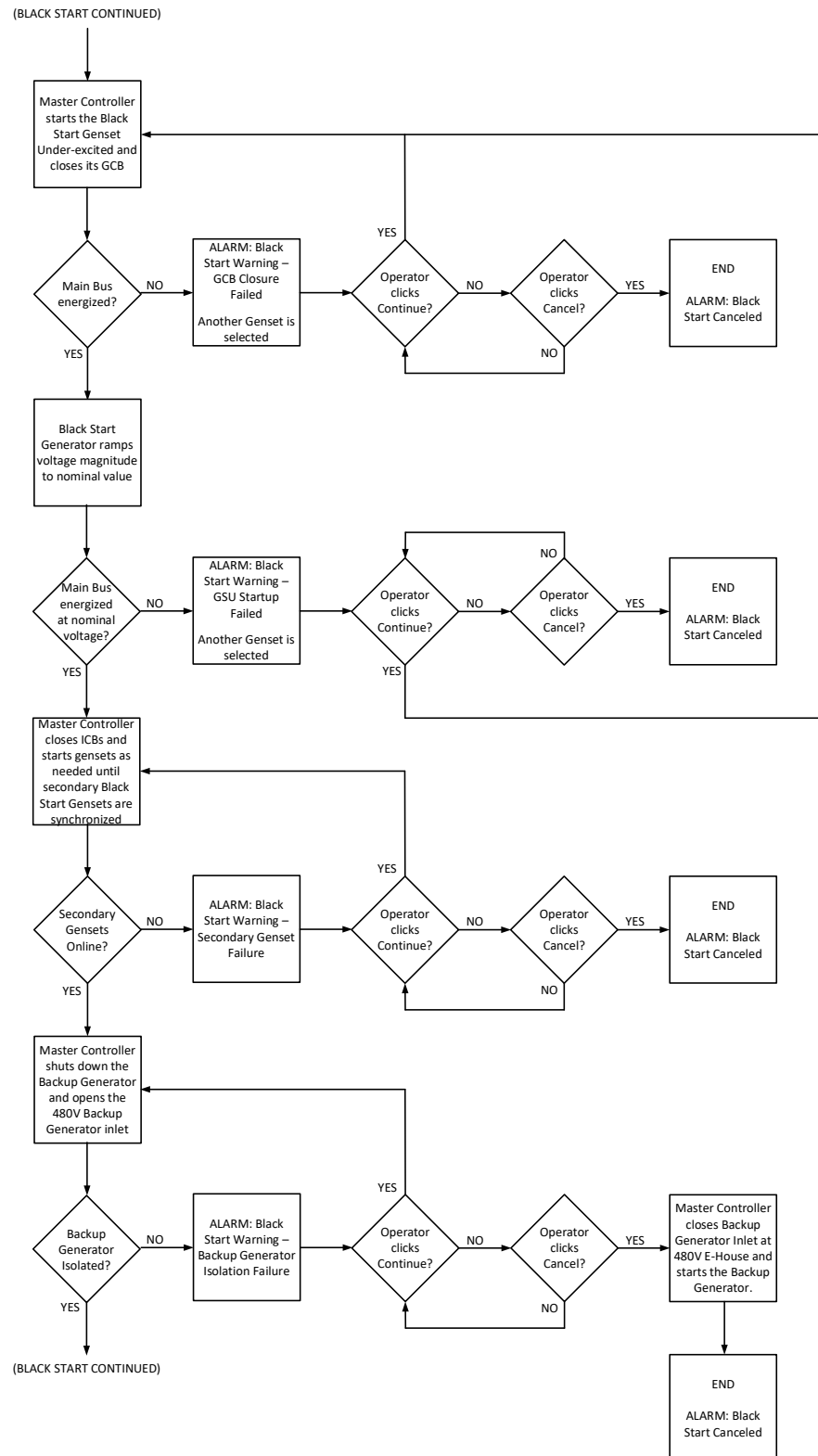


Figure 11 (continued): Black Start Sequence of Operations

(BLACK START CONTINUED)

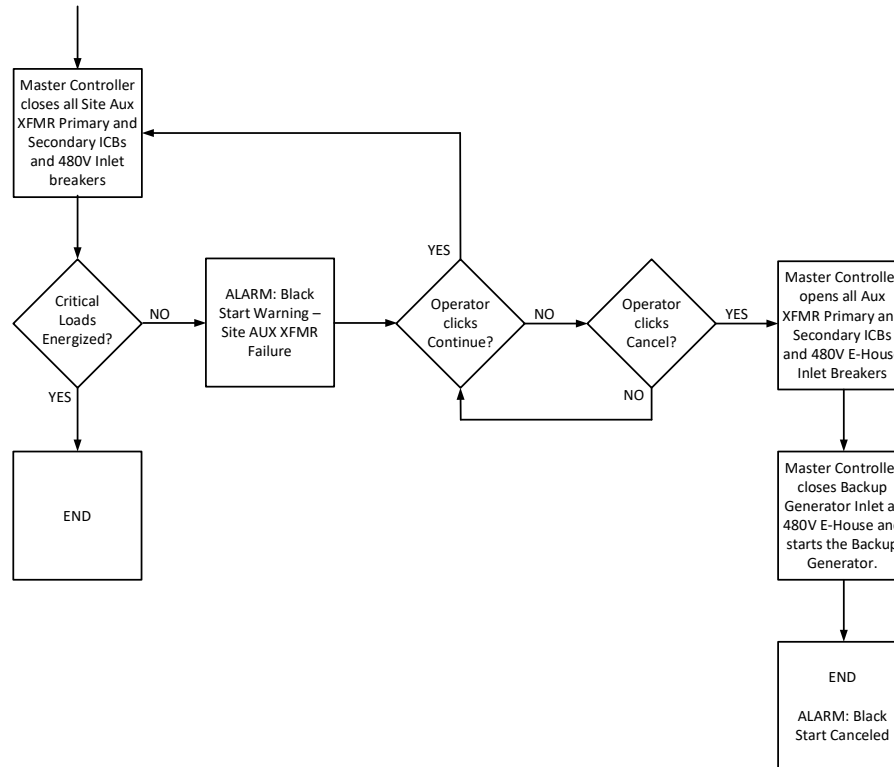


Figure 11 (continued): Black Start Sequence of Operations

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15.17. HISTORICAL DATA POINTS AND ALARMS

At any given point in time the Master Controller should make all data available to the OT Network that is required to completely reconstruct all plant equipment statuses. This data should include but is not limited to:

- breaker open/close states
- generator run/stop statuses
- pressures
- temperatures
- levels
- system power/voltage/current/frequency measurements
- alarms
- Master Controller control input/output states
- Speed/Voltage bias control signal values

All Master Controller Alarms/Alerts, internal variables, control outputs, feedback signal measurements, and other pertinent data should be provided to the VoltaGrid Message Bus and Control Room HMI via Modbus, or other Open Source Ethernet Protocol messages on the OT Network at all times.

15.18. FAILURE MODES

Master Controller Failures shall not result in system-wide blackouts.

The system shall include two redundant Master Controllers. The Secondary Master Controller shall monitor the health of the Primary Master Controller and take control if a Primary Controller health issue is detected.

If both the Primary and Secondary Master Controllers have failed and/or lost power, the system should shift to voltage and frequency Droop and all closed breakers/switches shall remain closed unless an electrical fault condition is detected by an electrical protection relay.

The Control Room HMI shall not be dependent on the Master Controller. If the Primary and Secondary Master Controllers have failed, the operator should be informed via alarm(s) but should still be able to see all relevant system information.

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16. OPERATIONAL TECHNOLOGY NETWORK

The Plant Control System shall include redundant OT Network equipment with robust communications routing capable of maintaining the continuous, uninterrupted flow of telemetric data and control signals between control system components (IEDs, Master Controllers, etc.). A typical Communications and Control Topology is shown below in **Figure 12**.

Note: Devices not critical to the Master Controller Interfaces such as LV ATS, LV Beaker Controllers, UPS, Battery Chargers, and synchronous machine controllers outside of the main 13.8kV Genset/DER controllers are not shown. These devices are also connected to the network switches in the Control and Communications topology shown below.

Note: Only one QPAC is shown, and only one of each type of device connected to a pair of switches is shown. For example, in an exhaustive Communications and Control Architecture Diagram, a J624 QPAC would have 5x SEL 700Gs connected to the QPAC Network Switches and there would be at least 4x SEL 751s connected to the FCB network switches.

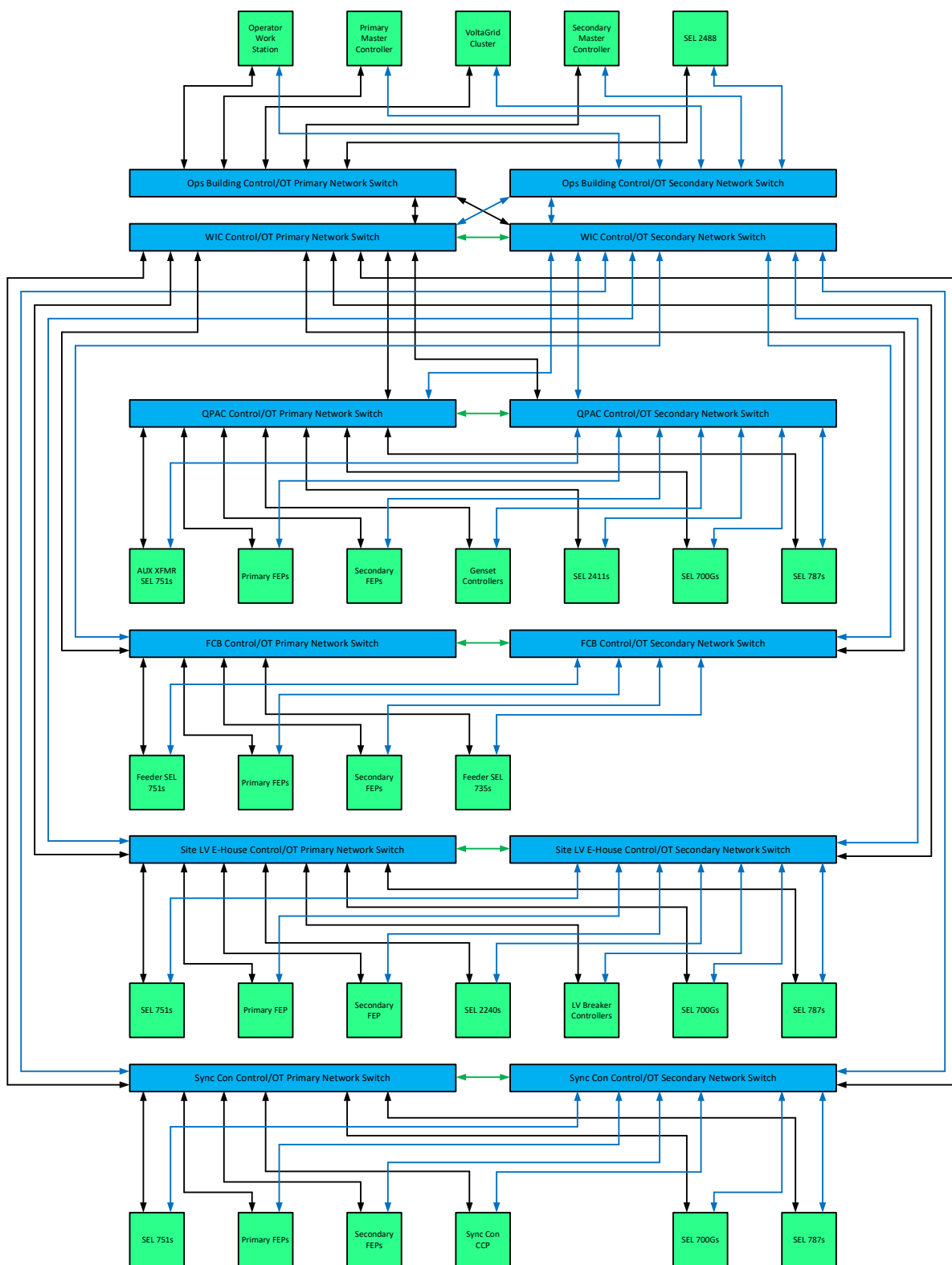


Figure 12: Typical Communications and Control Topology

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16.1. IP ADDRESSING

The OT Network Architecture shall support unique IP Addresses for each IED, PC, PLC, etc. that is connected to the OT Network.

Site specific IP Address Maps will be provided by VoltaGrid.

16.2. COMMUNICATIONS PROTOCOLS

The OT Networks shall provide each device with two ethernet ports – one for each of the two redundant OT network switches – capable of supporting the following communications protocols:

- Modbus
- IEC 61850 GOOSE
- IEC 61850 MMS
- SNMP
- NGVL
- IEEE C37.188
- SEL Protocol
- DNP3
- MQTT

16.3. CYBER SECURITY

The OT Network design and operation shall satisfy the requirements defined by ISO 27000.

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16.4. THROUGHPUT AND LATENCY

The OT Network shall be designed to the following specifications (at a minimum) to ensure seamless, continuous, low-latency control and monitoring of data transfer between devices and applications:

- 1GB/s Throughput
- 10GB/s Routing Throughput
- Less than 10ms Latency
- Less than 4ms Latency for GOOSE communications

16.5. TIME SYNCHRONIZATION

All PCs, PLCs, IEDs, servers, and applications which make up the Power Plant Control System shall be time synchronized via one of the following methods:

- Local connection and synchronization to a GNSS Antenna and IRIG-B Time Synchronization Clock (SEL 9524 and SEL 2401 or equivalent).
- Ethernet based PTP or NTP synchronization over the OT Network to a Central Time Synchronization Clock (SEL 2488 or equivalent).

The Central Time Synchronization Clock shall be connected to a GNSS antenna and synchronize its PTP and NTP time sources to its IRIG-B time source (GNSS Antenna and SEL 2488).

16.6. FAILURE MODES

All losses of communication, switch failures, port failures, and any other detectable OT Network health issues shall be exposed to the VoltaGrid Message Bus and Control Room HMI as OT Network Alarms.

Single switch, port, fiber, cable, or connector failures shall not result in any form of reduced Power Plant operability or visibility.

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17. VOLTAGRID CLUSTER

The VoltaGrid Cluster is the on-site Kubernetes cluster that hosts critical Plant Control System applications (the VoltaGrid Message Bus, Control Room HMI, VoltaGrid Gateway Server, and SWM). Kubernetes provides high-availability for these containerized applications by running them on redundant compute hardware and continuously reconciling actual system state with the desired system state.

If a node or container fails, the cluster will reschedule the affected Pods on other available nodes. Network services and Kubernetes Service objects then distribute traffic to the healthy Pods, restoring application availability and balancing load across the remaining compute resources.

In plain terms – if a single server goes down, the cluster automatically moves the applications that it was running to another server so that operators notice little or no disruption.

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17.1. VOLTAGRID MESSAGE BUS

The VoltaGrid Message Bus is a group of applications which collect data from all communications and control equipment on the OT Network and converts that data to MQTT so that it can be accessed, via the VoltaGrid Gateway Server, by VoltaGrid cloud storage and the Control Room HMI.

The VoltaGrid Message Bus will typically include JSON point lists (including Alarms/Alerts) for the following devices:

- Fortinet Network Switches, Firewalls, and other communications equipment
 - Communications interface with VoltaGrid Message Bus is TBD
- SEL 3555 Master Controller
 - Modbus Register Map to be provided by SEL
- SEL 3350 Genset FEP
 - Modbus Register Map to be provided by SEL
- INNIO DIANE Genset Controller
 - Modbus Register Map to be provided by INNIO
 - VoltaGrid to confirm that all pertinent data from engine, excitation, and AUX system controllers is aggregated in the Genset Controller Modbus Register Map
- SEL 751 or SEL 751A MV Genset AUX XFMR Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 751002B1B6X7B85BFC1
 - Part Number: 751A02B1B6X73850631
- SEL 2411 MV Genset Hardwired Interface Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 241122B1B1B746X0641
- SEL 700G MV Genset Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 0700G12B1B6X7682A6C1
- SEL 787 GSU Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 07873EE2B1B6X7685A6C1
- SEL 751 Customer Feeder Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 751002B1B6X7B85BFC1

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- SEL 735 Customer Feeder Power Meter
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 0735LX20921EFXB1XX16302EX
- SEL 2488 Time Synchronization Clock
 - Communications interface with VoltaGrid Message Bus is TBD
- SEL 787 Site AUX XFMR Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 07873EE2B1B6X7685A6C1
- SEL 2240 Site LV E-House Controller
 - Modbus Register Map to be provided by NSA
- Site LV E-House Breakers
 - Communications interface with VoltaGrid Message Bus is TBD – confirm with NSA
 - VoltaGrid Message Bus interface may be provided by the Site LV E-House Controller Modbus Server
- SEL 751 Site LV E-House Feeder Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 751002B1B6X7B85BFC1
- SEL 700G Site LV E-House Tie Breaker Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 0700G12B1B6X7682A6C1
- EMCP 4.4 Diesel Backup Genset Controller
 - Modbus Register Map to be provided by Caterpillar
- HUG SCR Controller
 - Modbus Register Map to be defined by HUG or INNIO
 - VoltaGrid to confirm with HUG that data from all IEDs in the SCR design will be aggregated in the Main SCR Controller or the Genset Controller Modbus Register Map
- Algas Gas Distribution Skid Controller
 - Modbus Register Map to be provided by Algas
 - VoltaGrid to confirm with Algas that data from all IEDs in the skid design will be aggregated in the Main Controller Modbus Register Map

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- Bulk Urea Distribution System Controller
 - Modbus Register Map to be provided by HUG
 - VoltaGrid to confirm with HUG that data from all IEDs in the Bulk Urea Distribution System will be aggregated in the Main Controller Modbus Register Map
- SEL 2240 BOP Remote I/O Module
 - Modbus Register Map to be defined by VoltaGrid
- LV ATS Controllers
 - Communications interface with VoltaGrid Cluster is TBD
 - Quantity, suppliers, and models to be determined
- 120VAC UPS (various models)
 - Communications interface with VoltaGrid Cluster is TBD
 - Quantity, suppliers, and models to be determined
- ABB AMS 1400 Synchronous Condenser CCP
 - Mobus Register Map provided by ABB in “4JUS000851P0084-A CCP Signal List”
 - Availability of data regarding Pony Motor VFD and MCC 3 BESS to be confirmed by VoltaGrid
 - VoltaGrid to confirm with ABB that data from all IEDs (including AUX system data) in the Sync Con system will be aggregated in the CCP Modbus Register Map
- ABB AMS 1400 LV E-House Breakers
 - Communications interface with VoltaGrid Message Bus is TBD
 - VoltaGrid Message Bus interface may be provided by the Sync Con CCP
- SEL 700G Synchronous Condenser Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 0700G12B1B6X7682A6C1
- SEL 751 Synchronous Condenser E-House Feeder Protection Relay
 - Mobus Register Map provided in SEL Relay Instruction Manual
 - Part Number: 751002B1B6X7B85BFC1

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17.2. ALARM/ALERT MANAGEMENT

All Alarms and Alerts, from all communications and control equipment, should be collected and organized into a uniform format by the VoltaGrid Cluster/Message Bus.

Regardless of how a particular Alarm/Alert is classified/transmitted by the equipment/IED OEM that defined and/or generated it, the VoltaGrid Message Bus shall categorize them as follows:

- Alarm (Red) – An equipment failure or condition will cause, or has caused, one or more pieces of equipment to turn off, deenergize, open (breakers), close (valves), or otherwise become inoperable.
 - A breaker has opened due to a short circuit.
 - An engine has shut down due to high jacket water temperature.
 - A Fuel Gas Distribution System ESDV has opened due to a high pressure condition.
 - An IED or Network Switch is in a failed state and is no longer communicating.
- Alert (Yellow) – The state of the system or a piece of equipment has changed in a way that may be pertinent to the Operator.
 - A measurement for a system pressure, temperature, flow rate, etc. has entered an abnormal range but no equipment shutdown is imminent.
 - A feedback signal from an IED is missing but no equipment shutdown is imminent.
 - Ethernet Communication was temporarily lost with an IED or group of IEDs but no equipment shutdown is imminent.

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All Alarms/Alerts should contain the following information at a minimum:

- A unique title
 - “High Jacket Water Temperature”
 - “Instantaneous Overcurrent”
- The name and location of the IED which generated the Alarm/Alert
 - “Genset Unit 76 Genset Controller”
 - “Genset Unit 132 XFMR Protection Relay”
- A detailed description of the event
 - “Engine jacket water temperature has exceeded safe operating levels, and the engine has been shut down.”
 - “Current through the ICB has exceeded safe levels, and the breaker has opened.”
- Recommended corrective actions (Alarms Only)
 - “Inspect the engine cooling water pump(s) and associated 480V power distribution systems.”
 - “Inspect the MV breakers, XFMRs, cabling, and terminations associated with unit 132.”
- Severity level (Alarms Only)
 - Level 1 – The failure can cause, or has caused, a system-wide black out
 - Example: An ICB failed to open after an Overcurrent Trip was asserted
 - Level 2 – Equipment has been shut down and equipment is likely to have been damaged
 - Example: An Overcurrent Trip was asserted on an electrical protection relay
 - Level 3 – Equipment shut down has been initiated as equipment damage was imminent
 - Example: An Undervoltage Trip was asserted on an electrical protection relay

The exhaustive list and details regarding VoltaGrid Data Center Power Plant Alarms/Alerts will be maintained separately from this specification.

Significant efforts should be made to eliminate nuisance Alarms/Alerts. For example, if an SEL 700G temporarily asserts an undervoltage pickup during an engine start, this should be recognized as normal behavior and no Alarm or Alert should be presented to the Operator.

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17.3. CUSTOMER INTERFACE

Data Centers and Data Center Electrical Distribution Systems should include a plurality of telemetric devices and forward looking notification signals which may be of use to the Plant Control System. The VoltaGrid Cluster and Message Bus shall be capable of receiving these data points as defined by VoltaGrid. These items may include but are not limited to the following:

- Electrical Distribution Voltage/Frequency Feedback Signals
- Electrical Distribution Breaker Status Feedback Signals
- Load Cycle Start Notification Signals
- Load Cycle Stop Notification Signals
- Fire Alarm Signals
- Emergency Shut Down Signals

Any data points used by the Master Controller shall be exposed to the OT Network via an Open Source Ethernet Protocol.

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17.4. CONTROL ROOM HUMAN MACHINE INTERFACE

VoltaGrid will design a Control Room HMI using Inductive Automation's Ignition platform to satisfy the requirements outlined in this specification. The Control Room HMI application will be served up by the VoltaGrid Cluster and shall be accessible locally via a web browser on the local OT Network and remotely via the VoltaGrid Gateway Server. Efforts will be made to configure the Control Room HMI with Perspective such that it is remotely accessible from cell phones as well as PCs.

All control outputs from the Control Room HMI shall be initiated and confirmed by an Operator, then sent to the Master Controller over an Open Source Ethernet Protocol. The Master Controller will execute control requests from the Operator provided that the plant conditions indicate that the action is safe and valid. If the Master controller is unable, for any reason, to execute an Operator's request, the Master Controller shall transmit an Alert on the OT Network via an Open Source Ethernet Protocol message.

All telemetry and alarm tags used in the development of the Control Room HMI shall be populated by the VoltaGrid Message Bus MQTT Broker.

The Master Controller OEM shall provide a soft I/O interface specification (preferably a Modbus Register Map) for Control Room HMI Operator Control Request Integration. See [Message Bus and Control Room HMI Interface](#).

Note: All Button/Selector clicks/presses shall require confirmation from the Operator before any Control functions are executed.

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The Control Room HMI shall consist of the following screens:

- Site Overview Screen— a 2D or 3D (future) site layout showing high level equipment statuses including load levels, flow rates, alarm states, and running/closed/open states
- MV PMS Screen – a Single Line Diagram showing mid-level detail regarding the MV Gensets, MV DERs, and the MV Electrical Distribution System
 - MV Genset Screen – Various values, depictions, selectors, and buttons required for the detailed review and control of a single genset
 - MV DER Screen – Various values, depictions, selectors, and buttons required for the detailed review and control of a single DER
 - MV Breaker Screen – Various values, depictions, selectors, and buttons required for the detailed review and control of a single MV breaker (and/or the asset(s) it is protecting)
 - If applicable, the Device Overview data/visualizations from SEL AcSElerator Quickset for the associated electrical protection relay should be shown here
- LV PMS Screen – a Single Line Diagram showing mid-level detail regarding the Backup Diesel Genset(s) and the LV Electrical Distribution System
 - Backup Diesel Genset Screen – Various alarms, values, depictions, selectors, and buttons required for the detailed review and control of a single Backup Diesel Genset
 - LV Breaker Screen – Various alarms, values, depictions, selectors, and buttons required for the detailed review and control of a single LV breaker (and/or the asset(s) it is protecting)
 - If applicable, the Device Overview data/visualizations from SEL AcSElerator Quickset for the associated electrical protection relay should be shown here
- BOPMS Screen – Fuel Gas Distribution System, Bulk Urea Distribution System, and SCR PFDs showing mid-level detail regarding Balance of Plant alarms, valve positions, tank levels, flow rates, pressures, temperatures, and dosing rates
- CCMS Screen – Network and Controls topology showing mid-level detail regarding the OT Network (switches, firewalls, patch panels, etc.) and controls equipment (Master Controllers, Genset Controllers, Remote I/O, FEPs, etc.)
- Alarm Management Screen – A table showing both current and historical Alerts and Alarms as well as their associated asset locations, descriptions, and corrective actions
- Time-Series Trending Screen – A time-series graph of plant data selected by the operator

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17.4.1. Site Overview Screen

The Site Overview Screen should show/do the following:

- Shows a 2D or 3D Site Layout of the Power Plant including all assets for which alarms and detailed screens are available
 - Assets should be color coded according to their health status
 - No Alarms/Alerts = Green
 - Active Alerts = Yellow
 - Active Alarms = Red
 - Clicking an asset should cause the HMI to navigate to the detailed asset screen
 - MV Genset Screen, LV Breaker Screen, Backup Diesel Genset Screen, etc.
 - Zooming in/out on this screen should show more less detailed telemetry for assets in view
- After a successful Blackout Recovery – Provides a Blackstart button which executes the Master Controllers Black Start Sequence
- Provides a “Party Mode” button which puts the Master Controller in Party Mode
- Shows a Navigation bar (bottom) allowing quick access to the MV PMS, LV PMS, BOPMS, CCMS, Alarm Management, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Shows an Alarm/Alert Banner (top)
 - Shows the most recent Alarm
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms
 - Includes a button which will navigate to the Alarm Management Screen

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17.4.2. MV Power Management Screen

The MV PMS Screen should show/do the following:

- Shows a 2D MV SLD for the Power Plant including all MV Gensets, MV DERs, MV SWGR, MV XFMRs and their associated bussing and statuses
 - At a minimum shows the real and reactive power levels, voltages, and frequencies at the following assets
 - MV Gensets
 - MV DERs
 - GCBs
 - ICBs
 - Main Bus/FCBs
 - Symbols should be color coded according to their running/stopped and open/closed states
 - Running/Closed/Energized = Red
 - Stopped/Open/Deenergized = Green
 - Starting/Synchronizing = Blue
 - Stopping/Unacknowledged Alarm = Blinking-Yellow and Red, Green, or Blue
 - A symbol should be included below each Genset Icon indicating its operating mode and alarm state
 - Blocked = Green Square
 - Auto = Green “Play Arrow”
 - Manual = Green Circle
 - Active Alarm = Blinking-Yellow Square, Play Arrow, or Circle
 - Clicking an asset should cause the HMI to navigate to the detailed asset screen
 - MV Genset Screen, MV DER Screen, MV Breaker Screen, etc.
 - Zooming in/out on this screen should show more less detailed telemetry for assets in view
- Shows a Navigation bar (bottom) allowing quick access to the Site Overview, LV PMS, BOPMS, CCMS, Alarm Management, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Shows an Alarm/Alert Banner (top)
 - Shows the most recent Alarm associated with the MV PMS System
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms associated with the MV PMS System
 - Includes a button which will navigate to the Alarm Management Screen

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17.4.3. MV Genset Screen

The MV Genset Screen should show/do the following:

- Shows Genset Run and Operating Mode states including the Operator Panel states from the local Genset Controller
- Shows critical Genset parameters including but not limited to:
 - Phase Voltages and Currents
 - Frequency
 - Real, Reactive, and Apparent Power
 - Power Factor
 - Engine Speed
 - Fuel Gas, Engine Oil, Exhaust, and Engine Cooling Water Pressures
 - Fuel Gas, Engine Oil, Exhaust, Turbo, SCR Inlet, Engine Cooling Water, and Return Water Temperatures
 - GCB open/closed status
- Shows local Genset control panel Operator Panel selections (see [Manual Backup Control Devices](#))
 - Operation Selection value
 - Request Selection Value
 - Synch Selection Value
- Provides the following Buttons/Selectors:
 - Block/Auto/Manual Selector
 - Block – the Genset is remotely blocked and starting from a local genset control panel should be disabled
 - Auto – Automatic starting for Master Controller IRM management is enabled, so long as remote control is enabled from the local genset control panel
 - Manual – Manual starting from the Control Room HMI is enabled, so long as remote control is enabled from the local genset control panel
 - **Note:** See [Manual Backup Control Devices](#)
 - Start Button
 - Stop Button
 - Alarm/Alert Acknowledgement Button
 - Genset Priority Level Selector/text box (1 to 100)
- Show Active Alarms and Alerts for the selected MV Genset

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17.4.4. MV DER Screen

The MV DER Screen should show/do the following:

- Show DER Run and Operating Mode states including the Operator Panel states from the local DER Controller
- Show critical DER parameters including but not limited to:
 - Phase Voltages and Currents
 - Frequency
 - Real, Reactive, and Apparent Power
 - Power Factor
 - **More parameters defined by VoltaGrid after design review**
- Provide the following Buttons/Selectors:
 - Block/Auto/Manual Selector
 - Start Button
 - Stop Button
 - Alarm/Alert Acknowledgement Button
- Show Active Alarms and Alerts for the selected MV DER

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17.4.5. MV Breaker Screen

The MV Breaker Screen should show/do the following:

- Show MV Breaker Open/Closed state
- Show critical MV Breaker parameters including but not limited to:
 - Phase Voltages and Currents
 - Frequency
 - Real, Reactive, and Apparent Power
 - Power Factor
- Show Protection Relay Statuses including but not limited to:
 - Digital I/O values
 - Analog I/O values
 - TRIP/CLOSE Logic
- Provide the following Buttons/Selectors:
 - Open Button
 - Close Button
 - Alarm/Alert Acknowledgement Button
- Show Active Alarms and Alerts for the selected MV Breaker

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17.4.6. LV Power Management Screen

The LV PMS Screen should show/do the following:

- Show a 2D MV SLD for the Power Plant including all LV Gensets, LV Breakers, LV XFMRs, LV ATS and their associated bussing and statuses
 - At a minimum shows the real and reactive power levels, voltages, and frequencies at the following assets
 - LV Gensets
 - Site LV E-House Breakers (where data is available)
 - Sync Con LV E-House Breakers (where data is available)
 - LV ATS (where data is available)
 - Symbols should be color coded according to their running/stopped and open/closed states
 - Running/Closed/Energized = Red
 - Stopped/Open/Deenergized = Green
 - Starting/Synchronizing = Blue
 - Stopping/Unacknowledged Alarm = Blinking-Yellow and Red, Green, or Blue
 - A symbol should be included below each Genset Icon indicating its operating mode and alarm state
 - Blocked = Green Square
 - Auto = Green “Play Arrow”
 - Manual = Green Circle
 - Active Alarm = Blinking-Yellow Square, Play Arrow, or Circle
 - Clicking an asset should cause the HMI to navigate to the detailed asset screen
 - Backup Diesel Genset Screen, LV Breaker Screen, etc.
 - Zooming in/out on this screen should show more less detailed telemetry for assets in view
- Show a Navigation bar (bottom) allowing quick access to the Site Overview, MV PMS, BOPMS, CCMS, Alarm Management, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Show an Alarm/Alert Banner (top)
 - Shows the most recent Alarm associated with the LV PMS System
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms associated with the LV PMS System
 - Includes a button which will navigate to the Alarm Management Screen

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17.4.7. Backup Diesel Genset Screen

The Backup Diesel Genset Screen should show/do the following:

- Show Genset Run and Operating Mode states including the Operator Panel states from the local Genset Controller
- Show critical Genset parameters including but not limited to:
 - Phase Voltages and Currents
 - Frequency
 - Real, Reactive, and Apparent Power
 - Power Factor
 - Engine Speed
 - Fuel Gas, Engine Oil, Exhaust, and Engine Cooling Water Pressures
 - Fuel Gas, Engine Oil, Exhaust, Turbo, Engine Cooling Water, and Return Water Temperatures
- Provide the following Buttons/Selectors:
 - Block/Auto/Manual Selector
 - Start Button
 - Stop Button
 - Alarm/Alert Acknowledgement Button
- Show Active Alarms and Alerts for the selected Backup Diesel Genset Screen

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17.4.8. LV Breaker Screen

The LV Breaker Screen should show/do the following:

- Show LV Breaker Open/Closed state
- Show critical LV Breaker parameters including but not limited to:
 - Phase Voltages and Currents
 - Frequency
 - Real, Reactive, and Apparent Power
 - Power Factor
- (Where applicable) Show Protection Relay Statuses including but not limited to:
 - Digital I/O values
 - Analog I/O values
 - TRIP/CLOSE Logic
- Provide the following Buttons/Selectors:
 - Open Button
 - Close Button
 - Alarm/Alert Acknowledgement Button
- Show Active Alarms and Alerts for the selected LV Breaker

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17.4.9. Balance of Plant Management Screen

The BOPMS Screen should show/do the following:

- Show 2D PFDs for the Fuel Gas and Bulk Urea Distribution Systems and their associated manifolds
 - At a minimum show the Levels, Pressures, Temperatures, Flow Rates, and Dosing Rates available at the following locations:
 - Main Fuel Gas Inlet
 - Algas Skid Outlets
 - Central Fuel Gas Manifold
 - Genset Fuel Gas Inlets
 - Bulk Urea Tanks
 - Urea Pump Skid Outlets
 - Central Urea Manifold
 - SCR Package Day Tanks
 - SCR Packages
 - Symbols should be color coded according to their open/closed states
 - Isolated/Closed/Not flowing = Green
 - Pressurized/Open/Flowing = Red
 - Starting/Available = Blue
 - Stopping/Closing/Unacknowledged Alarm = Blinking-Yellow and Red, Green, or Blue
 - Zooming in/out on this screen should show more less detailed telemetry for assets in view
- Show a Navigation bar (bottom) allowing quick access to the Site Overview, LV PMS, MV PMS, CCMS, Alarm Management, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Show an Alarm/Alert Banner (top)
 - Shows the most recent Alarm associated with the BOPMS System
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms associated with the BOPMS System
 - Includes a button which will navigate to the Alarm Management Screen

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17.4.10. Algas Skid Screen (FUTURE)

Requirements to be determined by VoltaGrid.

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17.4.11. Urea Pump Skid Screen (FUTURE)

Requirements to be determined by VoltaGrid.

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17.4.12. Control and Communication Management Screen

The CCMS Screen should show/do the following:

- Show a 2D Topology of the Plant OT Network and Controls architecture
 - At a minimum show the active/inactive and alarm states, Control voltage measurements, and UPS/Battery Charger statuses for the following devices:
 - VoltaGrid Cluster
 - All Network Switches
 - All Genset and SCR Controllers
 - All BOP System Controllers
 - All FEPs and Remote I/O Modules
 - All DER Controllers
 - All Electrical Protection Relays
 - Symbols should be color coded according to their active/inactive/alarm states
 - Active = Red
 - Inactive/Available = Green
 - Alarm = Blinking-Yellow and Red or Green
- Show a Navigation bar (bottom) allowing quick access to the Site Overview, LV PMS, MV PMS, BOPMS, Alarm Management, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Show an Alarm/Alert Banner (top)
 - Shows the most recent Alarm associated with the CCMS System
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms associated with the BOPMS System
 - Includes a button which will navigate to the Alarm Management Screen

17.4.13. Alarm Management Screen

The Alarm Management Screen should show/do the following:

- Show all Active Alarms and Alerts (when not hidden)
- Provide a toggle used to show/hide Alerts
- Provide a toggle used to show/hide historical Alarms and Alerts
- Provide Button(s) for acknowledging Alarms/Alerts
- Provide a Button to generate Alarm Report PDF/CSV files
- Show a Navigation bar (bottom) allowing quick access to the Site Overview, LV PMS, MV PMS, BOPMS, CCMS, and Time-Series Trending Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Show an Alarm/Alert Banner (top)
 - Shows the most recent Alarm
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms
 - Includes a button which will navigate to the Alarm Management Screen

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17.4.14. Time-Series Trending Screen

The Time-Series Trending Screen should show/do the following:

- Allow the Operator to select parameters from a list and display the selected parameters on a Time-Series Graph.
 - All MQTT Data available from any VoltaGrid Message Bus Point List should be accessible by the Time-Series Trending Screen
- Allow the Operator to save lists of preselected parameters to populate the Time-Series Trend
- Allow the Operator to place flags in the Time-Series Trend with notes associated to the flag
- Allow the Operator to zoom in/out and pan across the Time-Series Trend
- Provide a button for exporting Time-Series Trends as an image or CSV file
- Provide a toggle which will show/hide flags on the Time-Series Trend for Alarms/Alerts
- Show a Navigation bar (bottom) allowing quick access to the Site Overview, LV PMS, MV PMS, BOPMS, CCMS, and Alarm Management Screens
 - Clicking any of these should navigate the HMI to the associated Management Screen
- Show an Alarm/Alert Banner (top)
 - Shows the most recent Alarm
 - Includes a button for acknowledging Alerts/Alarms
 - Includes a drop down arrow which expands the banner to show the 10 most recent alarms
 - Includes a button which will navigate to the Alarm Management Screen

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17.5. VOLTAGRID GATEWAY SERVER

The VoltaGrid Gateway Server is the on-site bridge between the OT Network and VoltaGrid’s cloud infrastructure. It provides a secure access point for cloud-based applications to access the VoltaGrid Message Bus. The VoltaGrid Gateway Server also provides a secure path for remote access to the Control Room HMI, Genset HMIs, Relay AcSEerator Quickset interfaces, SWM, etc.

The VoltaGrid Gateway server allows remote Operations, Engineering, and Executive personnel to remotely record, evaluate, and troubleshoot equipment, operations, and plant performance.

17.6. TIME-SERIES TRENDING APPLICATIONS

In addition to the Time-Series Trending Screen provided by the Control Room HMI, the VoltaGrid Cluster shall also host an instance of SWM for viewing high resolution frequency, voltage, and current data from SEL electrical protection relays. The VoltaGrid Cluster should retain at least 30 days of data for SWM.

Note: proper installation of SEL 735 Power Metering relay firmware, GNSS antennas, and IRIG-B time synchronization clocks should allow for 3kHz sample rate CWM in SWM. This allows for in depth analysis of Harmonics and sub-transient system voltage/frequency performance.

17.7. FAILURE MODES

The failure modes of the VoltaGrid Cluster are defined by Kubernetes Cluster documentation. In general, a single server failure (or other VoltaGrid Cluster component failure) shall not result in a loss of Plant functionality, accessibility, operability, or visibility.

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18. TEST REPORTS

The following sections detail the test reports, from OEM and third party organizations, which are required to validate the Power Plant Control System.

18.1. COMPONENT FACTORY ACCEPTANCE TEST REPORTS

OEMs shall provide FAT Test Reports for all IEDs, Controllers, Relays, RTACs, switches, routers, firewalls, and all other control and communications equipment prior to shipping them to a VoltaGrid Data Center Power Plant location.

18.2. SYSTEM RTDS SIMULATION REPORT

The Master Controller and/or Electrical Protection Relay OEM(s) shall provide an RTDS Simulation Report to verify that the Gensets, DERs, and Plant Control System can satisfy the customer's voltage and frequency tolerances based on the anticipated customer load profile.

18.3. MASTER CONTROLLER HIL FACTORY ACCEPTANCE TEST REPORT

The Master Controller OEM shall provide a HIL FAT Report showcasing the Master Controller's ability to achieve all stated requirements found in the [Master Controller](#) section of this document.

The HIL FAT Report should also show system transient frequency/voltage deviation performance during all control operations and automatic control sequences defined in the [Master Controller](#) section of this document.

18.4. SITE ACCEPTANCE TEST REPORTS

All OEMs, Suppliers, and/or Integrators of any portion of the Plant Control System defined in this document shall perform on-site commissioning tests defined by a VoltaGrid Plant Commissioning Procedure and provide the results of those tests to VoltaGrid in the form of a SAT Report.